NEET

Time : 3 H 20 min
MM : 720

Solved Paper 2024

Instructions

- This question paper contains 200 Multiple Choice Question from the subjects **Physics, Chemistry** & **Biology** (Botany & Zoology). There are 50 questions in each subject, divided into two Sections : Section-A and Section B.
- **Section A** contains 35 questions in each subject (Q. Nos. 1 to 35, 51 to 85, 101 to 135 and 151 to 185) and all questions are compulsory.
- **Section B** contains 15 questions from each subject (Q. Nos. 36 to 50, 86 to 100, 136 to 150 and 186 to 200) out of which 10 questions are to be attempted.
- Each question carries 4 Marks. For each correct response, the candidate will get 4 marks. For each incorrect response 1 mark will be deducted from the total scores.

PHYSICS

Section-A

1. A bob is whirled in a horizontal plane by means of a string with an initial speed of ω rpm. The tension in the string is T. If speed becomes 2ω while keeping the same radius, the tension in the string becomes
(a) T (b) $4T$ (c) $T/4$ (d) $\sqrt{2}T$

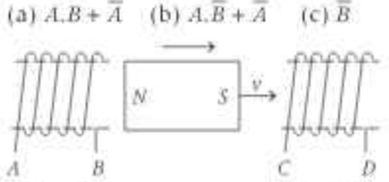
2. A particle moving with uniform speed in a circular path maintains
(a) constant velocity
(b) constant acceleration
(c) constant velocity but varying acceleration
(d) varying velocity and varying acceleration

3. A logic circuit provides the output Y as per the following truth table.

A	B	Y
0	0	1
0	1	0
1	0	1
1	1	0

The expression for the output Y is
(a) $A.B + \overline{A}$ (b) $A.\overline{B} + \overline{A}$ (c) $\overline{B}$ (d) B

4.



In the above diagram, a strong bar magnet is moving towards solenoid-2 from solenoid-1. The direction of induced current in solenoid-1 and that in solenoid-2, respectively, are through the directions.
(a) AB and DC (b) BA and CD
(c) AB and CD (d) BA and DC

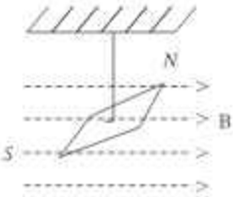
5. Given below are two statements. One is labelled as Assertion (A) and the other is labelled as Reason (R).
Assertion (A) The potential V at any axial point, at 2 m distance r from the centre of the dipole of dipole moment vector **p** of magnitude, 4×10^{-6} C m, is $\pm9\times10^{3}$ V.
(Take, $\frac{1}{4\pi\varepsilon_0} = 9\times10^{9}$ SI units)
Reason (R) $V = \pm\frac{2p}{4\pi\varepsilon_0 r^2}$, where r is the distance of any axial point, situated at 2 m from the centre of the dipole.
In the light of the above statements, choose the correct answer from the options given below.
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

6. Match List-I with List-II.

	List-I (Material)		**List**-II (Susceptibility (χ))
A	Diamagnetic	I.	$\chi = 0$
B	Ferromagnetic	II.	$0 > \chi \geq -1$
C	Paramagnetic	III.	$\chi >> 1$
D	Non-magnetic	IV.	$0 < \chi < \varepsilon$ (a small positive number)

Choose the correct answer from the options given below.
(a) A-II, B-III, C-IV, D-I (b) A-II, B-I, C-III, D-IV
(c) A-III, B-II, C-I, D-IV (d) A-IV, B-III, C-II, D-I

7. In a uniform magnetic field of 0.049 T, a magnetic needle performs 20 complete oscillations in 5 s as shown. The moment of inertia of the needle is 9.8×10^{-6} kg m^2. If the magnitude of magnetic moment of the needle is $x \times 10^{-5}$ Am^2, then the value of x is



(a) $5\pi^2$ (b) $128\pi^2$ (c) $50\pi^2$ (d) $1280\pi^2$

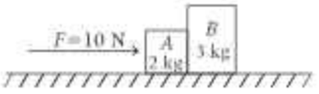
8. In an ideal transformer, the turns ratio is $\frac{N_p}{N_s} = \frac{1}{2}$. The ratio $V_s : V_p$ is equal to (the symbols carry their usual meaning)

(a) 1 : 2 (b) 2 : 1 (c) 1 : 1 (d) 1 : 4

9. In a vernier callipers, $(N + 1)$ divisions of vernier scale coincide with N divisions of main scale. If 1 MSD represents 0.1 mm, the vernier constant (in cm) is

(a) $\frac{1}{10N}$ (b) $\frac{1}{100(N+1)}$ (c) $100N$ (d) $10(N + 1)$

10. A horizontal force 10 N is applied to a block A as shown in figure. The mass of blocks A and B are 2 kg and 3 kg, respectively. The blocks slide over a frictionless surface. The force exerted by block A on block B is

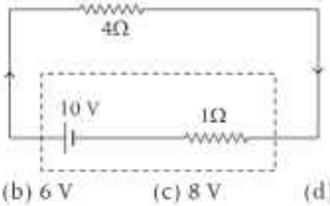


(a) zero (b) 4 N (c) 6 N (d) 10 N

11. If $x = 5 \sin\left(\pi t + \frac{\pi}{3}\right)$ m represents the motion of a particle executing simple harmonic motion, the amplitude and time period of motion, respectively, are

(a) 5 cm, 2 s (b) 5 m, 2 s (c) 5 cm, 1 s (d) 5 m, 1 s

12. The terminal voltage of the battery, whose emf is 10 V and internal resistance 1Ω when connected through an external resistance of 4 Ω as shown in the figure is



(a) 4 V (b) 6 V (c) 8 V (d) 10 V

13. Given below are two statements.

Statement I Atoms are electrically neutral as they contain equal number of positive and negative charges.

Statement II Atoms of each element are stable and emit their characteristic spectrum.

In the light of the above statements, choose the most appropriate answer from the options given below.

(a) Both Statement I and Statement II are correct.
(b) Both Statement I and Statement II are incorrect.
(c) Statement I is correct but Statement II is incorrect.
(d) Statement I is incorrect but Statement II is correct.

14. If c is the velocity of light in free space, the correct statements about photon among the following are

A. The energy of a photon is $E = h\nu$.
B. The velocity of a photon is c.
C. The momentum of a photon, $p = \frac{h\nu}{c}$
D. In a photon-electron collision, both total energy and total momentum are conserved.
E. Photon possesses positive charge.

Choose the correct answer from the options given below

(a) A and B only (b) A, B, C and D only
(c) A, C and D only (d) A, B, D and E only

15. Match List I with List II.

List I (Spectral lines of Hydrogen for transitions from)	List II (Wavelengths (nm))
A. $n_2 = 3$ to $n_1 = 2$	I. 410.2
B. $n_2 = 4$ to $n_1 = 2$	II. 434.1
C. $n_2 = 5$ to $n_1 = 2$	III. 656.3
D. $n_2 = 6$ to $n_1 = 2$	IV. 486.1

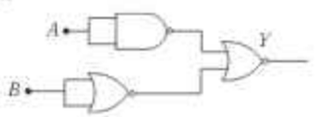
Choose the correct answer from the options given below.

(a) A-II, B-I, C-IV, D-III (b) A-III, B-IV, C-II, D-I
(c) A-IV, B-III, C-I, D-II (d) A-I, B-II, C-III, D-IV

16. A tightly wound 100 turns coil of radius 10 cm carries a current of 7 A. The magnitude of the magnetic field at the centre of the coil is (Take, permeability of free space as $4\pi \times 10^{-7}$ SI units)

(a) 44 mT (b) 4.4 T (c) 4.4 mT (d) 44 T

17. The output Y of the given logic gate is similar to the output of an/a



(a) NAND gate (b) NOR gate
(c) OR gate (d) AND gate

18. A wire of length l and resistance 100 Ω is divided into 10 equal parts. The first 5 parts are connected in series while the next 5 parts are connected in parallel. The two combinations are again connected in series. The resistance of this final combination is

(a) 26 Ω (b) 52 Ω (c) 55 Ω (d) 60 Ω

19. $^{290}_{82}X \xrightarrow{\alpha} {}^{286}_{80}Y \longrightarrow Z \xrightarrow{\beta^-} P \xrightarrow{e^-} Q$

In the nuclear emission stated above, the mass number and atomic number of the product Q respectively are

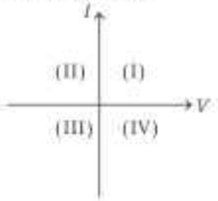
(a) 280, 81 (b) 286, 80 (c) 288, 82 (d) 286, 81

20. The maximum elongation of a steel wire of 1 m length if the elastic limit of steel and its Young's modulus, respectively, are 8×10^{8} Nm^{-2} and 2×10^{11} Nm^{-2}, is
(a) 4 mm (b) 0.4 mm (c) 40 mm (d) 8 mm

21. If the monochromatic source in Young's double slit experiment is replaced by white light, then
(a) interference pattern will disappear
(b) there will be a central dark fringe surrounded by a few coloured fringes
(c) there will be a central bright white fringe surrounded by a few coloured fringes
(d) all bright fringes will be of equal width

22. At any instant of time t, the displacement of any particle is given by $2t-1$ (SI unit) under the influence of force of 5 N. The value of instantaneous power is (in SI unit)
(a) 10 (b) 5 (c) 7 (d) 6

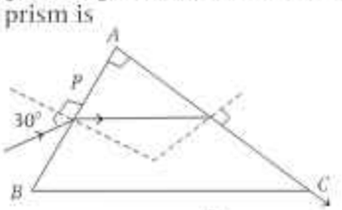
23. Consider the following statements A and B and identify the correct answer.



A. For a solar-cell, the I-V characteristics lies in the (IV) quadrant of the given graph.
B. In a reverse biased p-n junction diode, the current measured in (μA), is due to majority charge carriers.
(a) A is correct but B is incorrect.
(b) A is incorrect but B is correct.
(c) Both A and B are correct.
(d) Both A and B are incorrect.

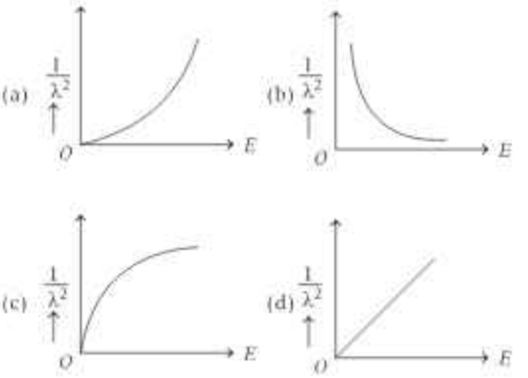
24. Two bodies A and B of same mass undergo completely inelastic one dimensional collision. The body A moves with velocity v_1 while body B is at rest before collision. The velocity of the system after collision is v_2. The ratio is
(a) 1 : 2 (b) 2 : 1 (c) 4 : 1 (d) 1 : 4

25. A light ray enters through a right angled prism at point P with the angle of incidence 30° as shown in figure. It travels through the prism parallel to its base BC and emerges along the face AC. The refractive index of the prism is



(a) $\frac{\sqrt{5}}{4}$ (b) $\frac{\sqrt{5}}{2}$
(c) $\frac{\sqrt{3}}{4}$ (d) $\frac{\sqrt{3}}{2}$

26. The graph which shows the variation of $\frac{1}{\lambda^2}$ and its kinetic energy, E is (where, λ is de-Broglie wavelength of a free particle)



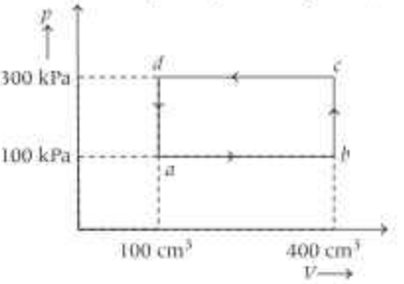
27. The quantities which have the same dimensions as those of solid angle are
(a) strain and angle
(b) stress and angle
(c) strain and arc
(d) angular speed and stress

28. An unpolarised light beam strikes a glass surface at Brewster's angle. Then
(a) the reflected light will be partially polarised.
(b) the refracted light will be completely polarised.
(c) both the reflected and refracted light will be completely polarised.
(d) the reflected light will be completely polarised but the refracted light will be partially polarised.

29. The moment of inertia of a thin rod about an axis passing through its mid-point and perpendicular to the rod is 2400 g cm^2. The length of the 400 g rod is nearly
(a) 8.5 cm (b) 17.5 cm
(c) 20.7 cm (d) 72.0 cm

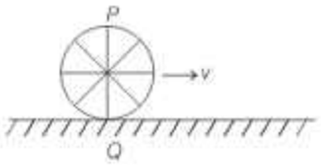
30. A thin flat circular disc of radius 4.5 cm is placed gently over the surface of water. If surface tension of water is 0.07 Nm^{-1}, then the excess force required to take it away from the surface is
(a) 19.8 mN (b) 198 N (c) 1.98 mN (d) 99 N

31. A thermodynamic system is taken through the cycle $abcda$. The work done by the gas along the path bc is



(a) zero (b) 30 J (c) −90 J (d) −60 J

32. A wheel of a bullock cart is rolling on a level road as shown in the figure below. If its linear speed is v in the direction shown, which one of the following options is correct (P and Q are any highest and lowest points on the wheel, respectively)?

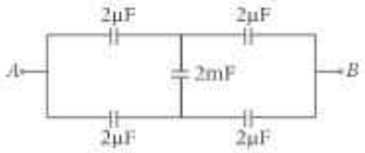


(a) Point P moves slower than point Q
(b) Point P moves faster than point Q
(c) Both the points P and Q move with equal speed
(d) Point P has zero speed

33. The mass of a planet is 1/10 th that of the earth and its diameter is half that of the earth. The acceleration due to gravity on that planet is

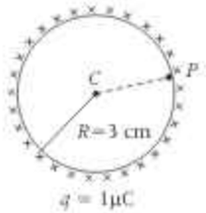
(a) 19.6 ms^{-2} (b) 9.8 ms^{-2}
(c) 4.9 ms^{-2} (d) 3.92 ms^{-2}

34. In the following circuit, the equivalent capacitance between terminal A and terminal B is



(a) 2 μF (b) 1 μF (c) 0.5 μF (d) 4 μF

35. A thin spherical shell is charged by some source. The potential difference between the two points C and P (in V) shown in the figure is

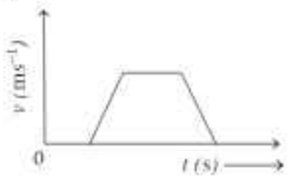


(Take, $\frac{1}{4\pi\varepsilon_0} = 9\times10^9$ SI units)

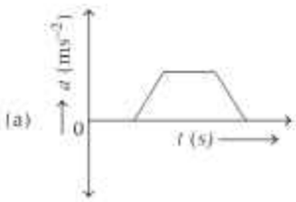
(a) 3×10^5 (b) 1×10^5
(c) 0.5×10^5 (d) zero

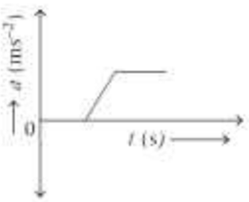
Section-B

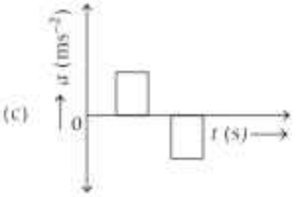
36. The velocity (v)-time (t) plot of the motion of a body is shown below

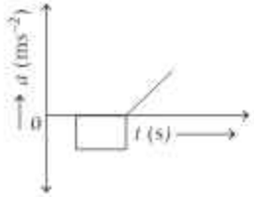


The acceleration (a)- time (t) graph that best suits this motion is

(a) 

(b) 

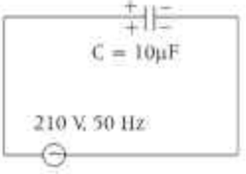
(c) 

(d) 

37. If the mass of the bob in a simple pendulum is increased to thrice its original mass and its length is made half its original length, then the new time period of oscillation is $x/2$ times its original time period. Then, the value of x is

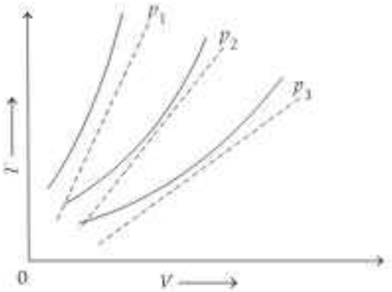
(a) $\sqrt{3}$ (b) $\sqrt{2}$ (c) $2\sqrt{3}$ (d) 4

38. A 10 μF capacitor is connected to a 210 V, 50 Hz source as shown in figure. The peak current in the circuit is nearly ($\pi = 3.14$).



(a) 0.58 A (b) 0.93 A
(c) 1.20 A (d) 0.35 A

39. The following graph represents the T-V curves of an ideal gas (where T is the temperature and V the volume) at three pressures p_1, p_2 and p_3 compared with those of Charles's law represented as dotted lines.



Then the correct relation is.

(a) $p_1 > p_2 > p_3$ (b) $p_1 > p_3 > p_2$
(c) $p_2 > p_1 > p_3$ (d) $p_1 > p_2 > p_3$

40. An iron bar of length L has magnetic moment M. It is bent at the middle of its length such that the two arms make an angle 60° with each other. The magnetic moment of this new magnet is

(a) M (b) $\frac{M}{2}$ (c) $2M$ (d) $\frac{M}{\sqrt{3}}$

41. The minimum energy required to launch a satellite of mass m from the surface of earth of mass M and radius R in a circular orbit at an altitude of $2R$ from the surface of the earth is

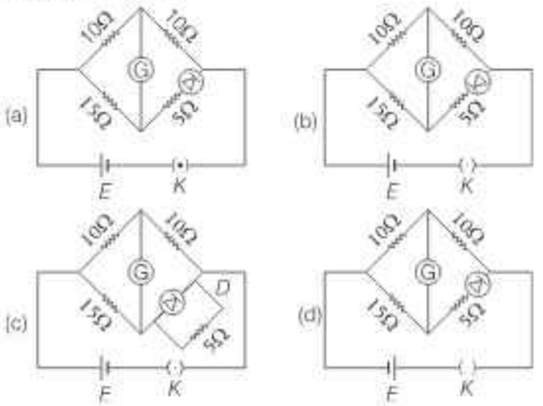
(a) $\frac{5GmM}{6R}$ (b) $\frac{2GmM}{3R}$ (c) $\frac{GmM}{2R}$ (d) $\frac{GmM}{3R}$

42. A parallel plate capacitor is charged by connecting it to a battery through a resistor. If I is the current in the circuit, then in the gap between the plates.
(a) There is no current.
(b) Displacement current of magnitude equal to I flows in the same direction as I.
(c) Displacement current of magnitude equal to I flows in a direction opposite to that of I.
(d) Displacement current of magnitude greater than I flows but can be in any direction.

43. The property which is not of an electromagnetic wave travelling in free space is that
(a) They are transverse in nature
(b) The energy density in electric field is equal to energy density in magnetic field
(c) They travel with a speed equal to $\frac{1}{\sqrt{\mu_0\varepsilon_0}}$
(d) They originate from charges moving with uniform speed

44. A metallic bar of Young's modulus, 0.5×10^{11} N m^{-2} and coefficient of linear thermal expansion $10^{-5}{}^{\circ}C^{-1}$, length 1 m and area of cross-section 10^{-3} m^2 is heated from 0°C to 100°C without expansion or bending. The compressive force developed in it is
(a) 5×10^3 N (b) 50×10^3 N
(c) 100×10^3 N (d) 2×10^3 N

45. Choose the correct circuit which can achieve the bridge balance.



46. A sheet is placed on a horizontal surface in front of a strong magnetic pole. A force is needed to
A. hold the sheet there if it is magnetic.
B. hold the sheet there if it is non-magnetic.
C. move the sheet away from the pole with uniform velocity if it is conducting.
D. move the sheet away from the pole with uniform velocity if it is both, non-conducting and non-polar.
Choose the correct statement(s) from the options given below
(a) B and D only (b) A and C only
(c) A, C and D only (d) C only

47. If the plates of a parallel plate capacitor connected to a battery are moved close to each other, then
A. the charge stored in it, increases.
B. the energy stored in it, decreases.
C. its capacitance increases.
D. the ratio of charge to its potential remains the same.
E. the product of charge and voltage increases.
Choose the most appropriate answer from the options given below
(a) A, B and E only (b) A, C and E only
(c) B, D and E only (d) A, B and C only

48. Two heaters A and B have power rating of 1 kW and 2 kW, respectively. Those two are first connected in series and then in parallel to a fixed power source. The ratio of power outputs for these two cases is
(a) 1 : 1 (b) 2 : 9 (c) 1 : 2 (d) 2 : 3

49. A small telescope has an objective of focal length 140 cm and an eyepiece of focal length 5.0 cm. The magnifying power of telescope for viewing a distant object is
(a) 34 (b) 28 (c) 17 (d) 32

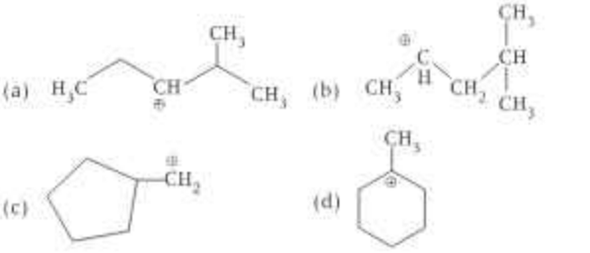
50. A force defined by $F=\alpha t^2+\beta t$ acts on a particle at a given time t. The factor which is dimensionless, if α and β are constants, is
(a) $\frac{\beta t}{\alpha}$ (b) $\frac{\alpha t}{\beta}$ (c) $\alpha\beta t$ (d) $\frac{\alpha\beta}{t}$

CHEMISTRY

Section-A

51. 'Spin only' magnetic moment is same for which of the following ions?
A. Ti^{3+} B. Cr^{2+}
C. Mn^{2+} D. Fe^{2+}
E. Sc^{3+}
Choose the most appropriate answer from the options given below.
(a) B and D only (b) A and E only
(c) B and C only (d) A and D only

52. The most stable carbocation among the following is



53. Given below are two statements:

Statement I The boiling point of hydrides of group 16 elements follow the order

$$H_2O > H_2Te > H_2Se > H_2S$$

Statement II On the basis of molecular mass, H_2O is expected to have lower boiling point than the other members of the group but due to the presence of extensive H-bonding in H_2O, it has higher boiling point.

In the light of the above statements, choose the correct answer from the options given below.

(a) Both Statement I and Statement II are true.
(b) Both Statement I and Statement II are false.
(c) Statement I is true but Statement II is false.
(d) Statement I is false but Statement II is true.

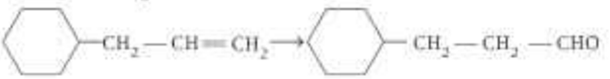
54. Match List I with List II.

	List I (Compound)		List II (Shape/geometry)
A.	NH_3	I.	Trigonal pyramidal
B.	BrF_5	II.	Square planar
C.	XeF_4	III.	Octahedral
D.	SF_6	IV.	Square pyramidal

Choose the correct answer from the options given below:
(a) A-I, B-IV, C-II, D-III (b) A-II, B-IV, C-III, D-I
(c) A-III, B-IV, C-I, D-II (d) A-II, B-III, C-IV, D-I

55. The highest number of helium atoms is in
(a) 4 moles of helium
(b) 4 u of helium
(c) 4 g of helium
(d) 2.271098 L of helium at STP

56. Identify the correct reagents that would bring about the following transformation.



(a) (i) $H_2O / H^{\oplus}$ (ii) CrO_3
(b) (i) BH_3 (ii) $H_2O_2 / \overset{\ominus}{O}H$
(iii) PCC
(c) (i) BH_3 (ii) $H_2O_2 / \overset{\ominus}{O}H$
(iii) Alk. $KMnO_4$ (iv) $H_3O^{\oplus}$
(d) (i) $H_2O / H^{\oplus}$ (ii) PCC

57. Match List I with List II.

	List-I (Process)		List-II (Conditions)
A.	Isothermal process	I.	No heat exchange
B.	Isochoric process	II.	Carried out at constant temperature
C.	Isobaric process	III.	Carried out at constant volume
D.	Adiabatic process	IV.	Carried out at constant pressure

Choose the correct answer from the options given below:
(a) A-IV, B-III, C-II, D-I
(b) A-IV, B-II, C-III, D-I
(c) A-I, B-II, C-III, D-IV
(d) A-II, B-III, C-IV, D-I

58. Which one of the following alcohols reacts instantaneously with Lucas reagent?

(a) $CH_3CH_2CH_2CH_2OH$ (b) $CH_3-CH_2-\underset{\displaystyle CH_3}{\underset{|}{CH}}-OH$

(c) $CH_3-\underset{\displaystyle CH_3}{\underset{|}{CH}}-CH_2OH$ (d) $CH_3-\underset{\displaystyle CH_3}{\overset{\displaystyle CH_3}{\overset{|}{\underset{|}{C}}}}-OH$

59. In which of the following equilibria, K_p and K_c are not equal?
(a) $PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$
(b) $H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$
(c) $CO(g) + H_2O(g) \rightleftharpoons CO_2(g) + H_2(g)$
(d) $2BrCl(g) \rightleftharpoons Br_2(g) + Cl_2(g)$

60. Match List I with List II.

	List I (Quantum Number)		List II (Information provided)
A.	m_l	I.	Shape of orbital
B.	m_s	II.	Size of orbital
C.	l	III.	Orientation of orbital
D.	n	IV.	Orientation of spin of electron

Choose the correct answer from the options given below :
(a) A-I, B-III, C-II, D-IV (b) A-III, B-IV, C-I, D-II
(c) A-III, B-IV, C-II, D-I (d) A-II, B-I, C-IV, D-III

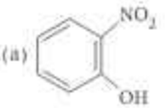
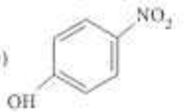
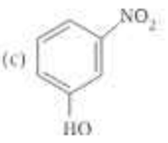
61. Given below are two statements.

Statement I Aniline does not undergo Friedel-Crafts alkylation reaction.

Statement II Aniline cannot be prepared through Gabriel synthesis.

In the light of the above statements, choose the correct answer from the options given below:
(a) Both statement I and statement II are true.
(b) Both statement I and statement II are false.
(c) Statement I is true but statement II is false.
(d) Statement I is false but Statement II is true.

62. Intramolecular hydrogen bonding is present in

(a) 
(b) 
(c) 
(d) HF

63. On heating, some solid substances change from solid to vapour state without passing through liquid state. The technique used for the purification of such solid substances based on the above principle is known as
(a) crystallisation (b) sublimation
(c) distillation (d) chromatography

64. In which of the following processes entropy increases?
A. A liquid evaporates to vapour.
B. Temperature of a crystalline solid lowered from 130 K to 0 K.
C. $2NaHCO_3(s) \longrightarrow Na_2CO_3(g) + CO_2(g) + H_2O(g)$
D. $Cl_2(g) \longrightarrow 2Cl(g)$
Choose the correct answer from the options given below:
(a) A and C (b) A, B and D
(c) A, C and D (d) C and D

65. Among group 16 elements, which one does not show –2 oxidation state?
(a) O (b) Se (c) Te (d) Po

66. Match List I with List II.

List I (Conversion)		List II (Number of Faraday required)
A. 1 mole of H_2O to O_2	I.	3F
B. 1 mole of MnO_4^- to Mn^{2+}	II.	2F
C. 1.5 mole of Ca from molten $CaCl_2$	III.	1F
D. 1 mole of FeO to Fe_2O_3	IV.	5F

Choose the correct answer from the options given below:
(a) A-II, B-IV, C-I, D-III (b) A-III, B-IV, C-I, D-II
(c) A-II, B-III, C-I, D-IV (d) A-III, B-IV, C-II, D-I

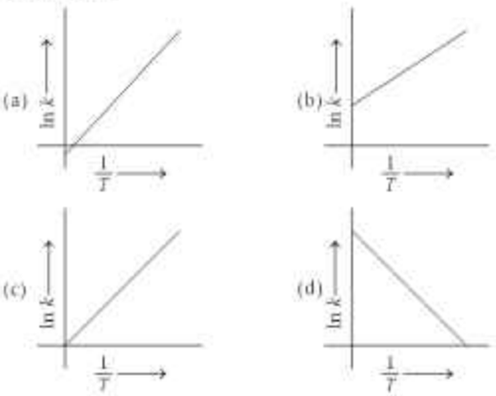
67. Arrange the following elements in increasing order of electronegativity:
N, O, F, C, Si
Choose the correct answer from the options given below:
(a) Si < C < N < O < F (b) Si < C < O < N < F
(c) O < F < N < C < Si (d) F < O < N < C < Si

68. A compound with a molecular formula of C_6H_{14} has two tertiary carbons. Its IUPAC name is
(a) *n*-hexane (b) 2-methylpentane
(c) 2,3-dimethylbutane (d) 2,2-dimethylbutane

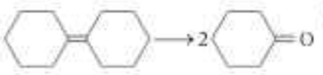
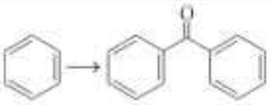
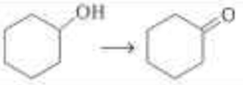
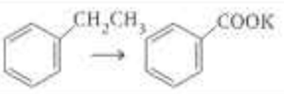
69. Fehling's solution '*A*' is
(a) aqueous copper sulphate
(b) alkaline copper sulphate
(c) alkaline solution of sodium potassium tartrate (Rochelle's salt)
(d) aqueous sodium citrate

70. Activation energy of any chemical reaction can be calculated if one knows the value of
(a) rate constant at standard temperature
(b) probability of collision
(c) orientation of reactant molecules during collision
(d) rate constant at two different temperatures

71. Which plot of ln k *vs* $1/T$ is consistent with Arrhenius equation?

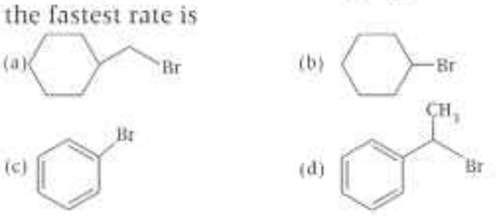


72. Match List I with List II.

List I (Reaction)		List II (Reagents/Condition)
A. 	I.	
B. 	II.	CrO_3
C. 	III.	$KMnO_4$ / KOH, Δ
D. 	IV.	(i) O_3 (ii) Zn - H_2O

Choose the correct answer from the options given below:
(a) A-IV, B-I, C-III, D-II (b) A-III, B-I, C-II, D-IV
(c) A-IV, B-I, C-II, D-III (d) A-I, B-IV, C-II, D-III

73. The compound that will undergo S_NI reaction with the fastest rate is



74. Which reaction is not a redox reaction?
(a) $Zn + CuSO_4 \longrightarrow ZnSO_4 + Cu$
(b) $2KClO_3 + I_2 \longrightarrow 2KIO_3 + Cl_2$
(c) $H_2 + Cl_2 \longrightarrow 2HCl$
(d) $BaCl_2 + Na_2SO_4 \longrightarrow BaSO_4 + 2NaCl$

75. Given below are two statements:
Statement I The boiling point of three isomeric pentanes follows the order
n-pentane > isopentane > neopentane
Statement II When branching increases, the molecule attains a shape of sphere. This results in smaller surface

area for contact, due to which the intermolecular forces between the spherical molecules are weak, thereby lowering the boiling point.

In the light of the above statements, choose the most appropriate answer from the options given below:
(a) Both Statement I and Statement II are true.
(b) Both Statement I and Statement II are false.
(c) Statement I is true but Statement II is false.
(d) Statement I is false but Statement II is true.

76. Given below are two statements :
Statement I Both $[Co(NH_3)_6]^{3+}$ and $[CoF_6]^{3-}$ complexes are octahedral but differ in their magnetic behaviour.
Statement II $[Co(NH_3)_6]^{3+}$ is diamagnetic whereas $[CoF_6]^{3-}$ is paramagnetic.

In the light of the above statements, choose the correct answer from the options given below:
(a) Both Statement I and Statement II are true.
(b) Both Statement I and Statement II are false.
(c) Statement I is true but Statement II is false.
(d) Statement I is false but Statement II is true.

77. Match List I with List II.

	List I (Molecule)		List II (Number and types of bond/s between two carbon atoms)
A.	Ethane	I.	One σ-bond and two π-bonds
B.	Ethene	II.	two π-bonds
C.	Carbon molecule, C_2	III.	One σ-bond
D.	Ethyne	IV.	One σ-bond and one π-bond

Choose the correct answer from the options given below:
(a) A-I, B-IV, C-II, D-III (b) A-IV, B-III, C-II, D-I
(c) A-III, B-IV, C-II, D-I (d) A-III, B-IV, C-I, D-II

78. The Henry's law constant (K_H) values of three gases (A, B, C) in water are 145, 2×10^{-5} and 35 kbar, respectively. The solubility of these gases in water follow the order:
(a) $B > A > C$ (b) $B > C > A$
(c) $A > C > B$ (d) $A > B > C$

79. The energy of an electron in the ground state ($n = 1$) for He^+ ion is –x J, then that for an electron in $n = 2$ state for Be^{3+} ion in J is
(a) $-x$ (b) $-\frac{x}{9}$ (c) $-4x$ (d) $-\frac{4}{9}x$

80. The E° value for the Mn^{3+}/Mn^{2+} couple is more positive than that of Cr^{3+}/Cr^{2+} or Fe^{3+}/Fe^{2+} due to change of
(a) d^5 to d^4 configuration (b) d^5 to d^2 configuration
(c) d^4 to d^5 configuration (d) d^3 to d^5 configuration

81. The reagents with which glucose does not react to give the corresponding tests/products are
A. Tollen's reagent B. Schiff's reagent
C. HCN D. NH_2OH
E. $NaHSO_3$
Choose the correct options from the given below:
(a) B and C (b) A and D
(c) B and E (d) E and D

82. Match List I with List II.

	List I (Complex)		List II (Type of isomerism)
A.	$[Co(NH_3)_5(NO_2)]Cl_2$	I.	Solvate isomerism
B.	$[Co(NH_3)_5(SO_4)]Br$	II.	Linkage isomerism
C.	$[Co(NH_3)_6][Cr(CN)_6]$	III.	Ionisation isomerism
D.	$[Co(H_2O)_6]Cl_3$	IV.	Coordination isomerism

Choose the correct answer from the options given below:
(a) A-II, B-III, C-IV, D-I (b) A-I, B-III, C-IV, D-II
(c) A-I, B-IV, C-III, D-II (d) A-II, B-IV, C-III, D-I

83. Arrange the following elements in increasing order of first ionisation enthalpy:
Li, Be, B, C, N
Choose the correct answer from the options given below:
(a) Li < Be < B < C < N (b) Li < B < Be < C < N
(c) Li < Be < C < B < N (d) Li < Be < N < B < C

84. 1 g of sodium hydroxide was treated with 25 mL of 0.75 M HCl solution, the mass of sodium hydroxide left unreacted is equal to
(a) 750 mg (b) 250 mg
(c) zero mg (d) 200 mg

85. For the reaction, $2A \rightleftharpoons B + C, K_c = 4\times10^{-3}$. At a given time, the composition of reaction mixture is: $[A] = [B] = [C] = 2\times10^{-3}$M.
Then, which of the following is correct?
(a) Reaction is at equilibrium.
(b) Reaction has a tendency to go in forward direction.
(c) Reaction has a tendency to go in backward direction.
(d) Reaction has gone to completion in forward direction.

Section-B

86. Given below are certain cations. Using inorganic qualitative analysis, arrange them in increasing group number from 0 to VI.
A. Al^{3+} B. Cu^{2+}
C. Ba^{2+} D. Co^{2+}
E. Mg^{2+}
Choose the correct answer from the options given below.
(a) B, A, D, C, E (b) B, C, A, D, E
(c) E, C, D, B, A (d) E, A, B, C, D

87. The products A and B obtained in the following reactions, respectively are

$$3ROH + PCl_3 \longrightarrow 3RCl + A$$
$$ROH + PCl_5 \longrightarrow RCl + HCl + B$$

(a) $POCl_3$ and H_3PO_3 (b) $POCl_3$ and H_3PO_4
(c) H_3PO_4 and $POCl_3$ (d) H_3PO_3 and $POCl_3$

88. Mass in grams of copper deposited by passing 9.6487 A current through a voltmeter containing copper sulphate solution for 100 seconds is (Given : Molar mass of Cu : 63 mol^{-1}, 1 F = 96487 C)
(a) 3.15 g (b) 0.315 g
(c) 31.5 g (d) 0.0315

89. The plot of osmotic pressure (π) *vs* concentration ($mol\ L^{-1}$) for a solution gives a straight line with slope 25.73 L bar mol^{-1}. The temperature at which the osmotic pressure measurement is done is
(Use, $R = 0.083$ L bar $mol^{-1}K^{-1}$)
(a) 37°C (b) 310°C (c) 25.73°C (d) 12.05°C

90. Identify the major product *C* formed in the following reaction sequence:

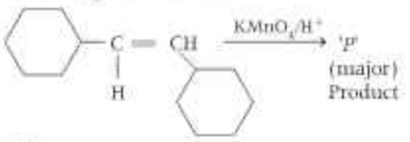
$$CH_3—CH_2—CH(I)—CH_3 \xrightarrow{NaCN} A \xrightarrow[\text{Partial hydrolysis}]{OH^-} B \xrightarrow[Br_2]{NaOH} C\ (\text{major})$$

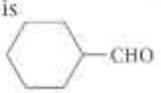
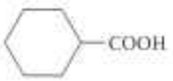
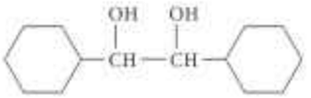
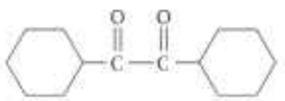
(a) propylamine (b) butylamine
(c) butanamide (d) α-bromobutanoic acid

91. Identify the correct answer.
(a) Three resonance structures can be drawn for ozone.
(b) BF_3 has non-zero dipole moment.
(c) Dipole moment of NF_3 is greater than that of NH_3.
(d) Three canonical forms can be drawn for CO_3^{2-} ion.

92. Given below are two statements :
Statement I $[Co(NH_3)_6]^{3+}$ is a homoleptic complex whereas $[Co(NH_3)_4Cl_2]^+$ is a heteroleptic complex.
Statement II Complex $[Co(NH_3)_6]^{3+}$ has only one kind of ligands but $[Co(NH_3)_4Cl_2]^+$ has more than one kind of ligands.
In the light of the above statements, choose the correct answer from the options given below.
(a) Both Statement I and Statement II are true.
(b) Both Statement I and Statement II are false.
(c) Statement I is true but Statement II is false.
(d) Statement I is false but Statement II is true.

93. For the given reaction



'P' is
(a) 
(b) 
(c) 
(d) 

94. The pair of lanthanoid ions which are diamagnetic is
(a) Ce^{4+} and Yb^{2+} (b) Ce^{3+} and Eu^{2+}
(c) Gd^{3+} and Eu^{3+} (d) Pm^{3+} and Sm^{3+}

95. Consider the following reaction in a sealed vessel at equilibrium with concentrations of
$N_2 = 3.0\times10^{-3}$ M, $O_2 = 4.2\times10^{-3}$ M
and $NO = 2.8\times10^{-3}$ M

$$2NO(g) \rightleftharpoons N_2(g) + O_2(g)$$

If 0.1 mol L^{-1} of NO (*g*) is taken in a closed vessel, what will be degree of dissociation (α) of NO(*g*) at equilibrium?
(a) 0.00889 (b) 0.0889 (c) 0.8889 (d) 0.717

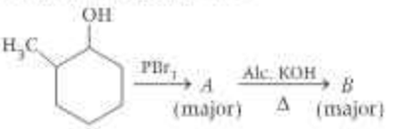
96. A compound *X* contains 32% of *A*, 20% of *B* and remaining percentage of *C*. Then, the empirical formula of *X* is
(Given, atomic masses of $A = 64; B = 40; C = 32$ u)
(a) A_2BC_2 (b) ABC_3 (c) AB_2C_2 (d) ABC_4

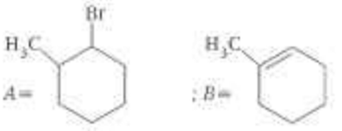
97. The work done during reversible isothermal expansion of one mole of hydrogen gas at 25°C from pressure of 20 atmosphere to 10 atmosphere is
(Given, $R = 2.0$ cal $K^{-1}mol^{-1}$)
(a) 0 calorie (b) −413.14 calories
(c) 413.14 calories (d) 100 calories

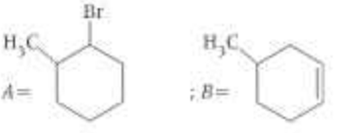
98. During the preparation of Mohr's salt solution (Ferrous ammonium sulphate), which of the following acid is added to prevent hydrolysis of Fe^{2+} ion?
(a) Dilute hydrochloric acid (b) Concentrated sulphuric acid
(c) Dilute nitric acid (d) Dilute sulphuric acid

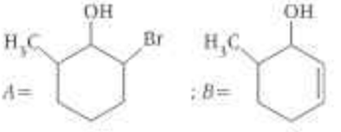
99. The rate of a reaction quadruples when temperature changes from 27°C to 57°C. Calculate the energy of activation.
Given, $R = 8.314$ J $K^{-1}mol^{-1}$, $\log 4 = 0.6021$
(a) 38.04 kJ/mol (b) 380.4 kJ/mol
(c) 3.80 kJ/mol (d) 3804 kJ/mol

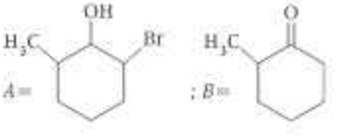
100. Major products *A* and *B* formed in the following reaction sequence are



(a) 

(b) 

(c) 

(d) 

BOTANY

Section-A

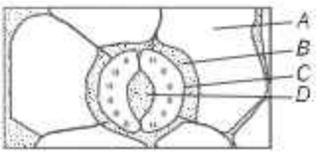
101. Lecithin, a small molecular weight organic compound found in living tissues, is an example of
(a) amino acids (b) phospholipids
(c) glycerides (d) carbohydrates

102. Which of the following are required for the dark reaction of photosynthesis?
A. Light B. Chlorophyll
C. CO_2 D. ATP
E. NADPH
Choose the correct answer from the options given below.
(a) A, B and C only (b) B, C and D only
(c) C, D and E only (d) D and E only

103. Spindle fibres attach to kinetochores of chromosomes during
(a) prophase (b) metaphase (c) anaphase (d) telophase

104. Bulliform cells are responsible for
(a) inward curling of leaves in monocots.
(b) protecting the plant from salt stress.
(c) increased photosynthesis in monocots.
(d) providing large spaces for storage of sugars.

105. In the given figure, which component has thin outer walls and highly thickened inner walls?



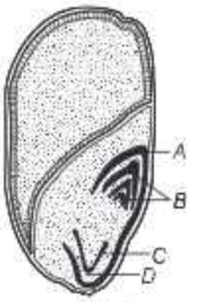
(a) C (b) D (c) A (d) B

106. What is the fate of a piece of DNA carrying only gene of interest which is transferred into an alien organism?
A. The piece of DNA would be able to multiply itself independently in the progeny cells of the organism.
B. It may get integrated into the genome of the recipient.
C. It may multiply and be inherited along with the host DNA.
D. The alien piece of DNA is not an integral part of chromosome.
E. It shows ability to replicate.
Choose the correct answer from the options given below.
(a) A and B only (b) D and E only
(c) B and C only (d) A and E only

107. Given below are two statements.
Statement I *Bt* toxins are insect group specific and coded by a gene *cry* IAc.
Statement II *Bt* toxin exists as inactive protoxin in *B. thuringiensis*. However, after ingestion by the insect the inactive protoxin gets converted into active form due to acidic pH of the insect gut.
In the light of the above statements, choose the correct answer from the options given below.
(a) Both Statement I and Statement II are true
(b) Both Statement I and Statement II are false
(c) Statement I is true but Statement II is false
(d) Statement I is false but Statement II is true

108. List of endangered species was released by
(a) GEAC (b) WWF (c) FOAM (d) IUCN

109. Identify the part of the seed from the given figure which is destined to form root when the seed germinates.



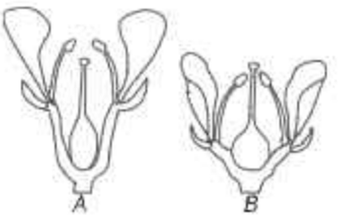
(a) A (b) B (c) C (d) D

110. Match List I with List II.

	List I		List II
A.	*Clostridium butylicum*	I.	Ethanol
B.	*Saccharomyces cerevisiae*	II.	Streptokinase
C.	*Trichoderma polysporum*	III.	Butyric acid
D.	*Streptococcus* sp.	IV.	Cyclosporin-A

Choose the correct answer from the options given below:
(a) A-III, B-I, C-II, D-IV (b) A-II, B-IV, C-III, D-I
(c) A-III, B-I, C-IV, D-II (d) A-IV, B-I, C-III, D-II

111. Identify the type of flowers based on the position of calyx, corolla and androecium with respect to the ovary from the given figures (A) and (B).



(a) (A) Epigynous; (B) Hypogynous
(b) (A) Hypogynous; (B) Epigynous
(c) (A) Perigynous; (B) Epigynous
(d) (A) Perigynous; (B) Perigynous

112. Auxin is used by gardeners to prepare weed-free lawns. But no damage is caused to grass as auxin
(a) promotes apical dominance.
(b) promotes abscission of mature leaves only.
(c) does not affect mature monocotyledonous plants.
(d) can help in cell division in grasses, to produce growth.

113. A pink flowered snapdragon plant was crossed with a red flowered Snapdragon plant. What type of phenotype/s is/are expected in the progeny?
(a) Only red flowered plants
(b) Red flowered as well as pink flowered plants
(c) Only pink flowered plants
(d) Red, pink as well as white flowered plants

114. Which one of the following is not a criterion for classification of fungi?
(a) Morphology of mycelium
(b) Mode of nutrition
(c) Mode of spore formation
(d) Fruiting body

115. The lactose present in the growth medium of bacteria is transported to the cell by the action of
(a) beta-galactosidase (b) acetylase
(c) permease (d) polymerase

116. In a plant, black seed colour (BB/Bb) is dominant over white seed colour (bb). In order to find out the genotype of the black seed plant, with which of the following genotype will you cross it?
(a) BB (b) bb
(c) Bb (d) BB/Bb

117. Given below are two statements.
Statement I Parenchyma is living, but collenchyma is dead tissue.
Statement II Gymnosperms lack xylem vessels, but presence of xylem vessels is the characteristic of angiosperms.
In the light of the above statements, choose the correct answer from the options given below.
(a) Both Statement I and Statement II are true
(b) Both Statement I and Statement II are false
(c) Statement I is true but Statement II is false
(d) Statement I is false but Statement II is true

118. How many molecules of ATP and NADPH are required for every molecule of CO_2 fixed in the Calvin cycle?
(a) 2 molecules of ATP and 3 molecules of NADPH
(b) 2 molecules of ATP and 2 molecules of NADPH
(c) 3 molecules of ATP and 3 molecules of NADPH
(d) 3 molecules of ATP and 2 molecules of NADPH

119. A transcription unit in DNA is defined primarily by the three regions in DNA and these are with respect to upstream and down stream end;
(a) Repressor, operator gene, structural gene
(b) Structural gene, transposons, operator gene
(c) Inducer, repressor, structural gene
(d) Promoter, structural gene, terminator

120. Tropical regions show greatest level of species richness because
A. tropical latitudes have remained relatively undisturbed for millions of years, hence more time was available for species diversification.
B. tropical environments are more seasonal.
C. more solar energy is available in tropics.
D. constant environments promote niche specialisation.
E. tropical environments are constant and predictable.
Choose the correct answer from the options given below.
(a) A, C, D and E only (b) A and B only
(c) A, B and E only (d) A, B and D only

121. The equation of Verhulst-Pearl logistic growth is $\frac{dN}{dt} = rN\left[\frac{K-N}{K}\right]$.
From this equation, K indicates
(a) intrinsic rate of natural increase
(b) biotic potential
(c) carrying capacity
(d) population density

122. Inhibition of succinic dehydrogenase enzyme by malonate is a classical example of
(a) cofactor inhibition (b) feedback inhibition
(c) competitive inhibition (d) enzyme activation

123. Which one of the following can be explained on the basis of Mendel's law of dominance?
A. Out of one pair of factors one is dominant and the other is recessive.
B. Alleles do not show any expression and both the characters appear as such in F_2-generation.
C. Factors occur in pairs in normal diploid plants.
D. The discrete unit controlling a particular character is called factor.
E. The expression of only one of the parental characters is found in a monohybrid cross.
Choose the correct answer from the options given below.
(a) A, B and C only (b) A, C, D and E only
(c) B, C and D only (d) A, B, C, D and E

124. Match List I with List II.

	List-I		List-II
A.	Nucleolus	I.	Site of formation of glycolipid
B.	Centriole	II.	Organisation like the cartwheel
C.	Leucoplasts	III.	Site for active ribosomal RNA synthesis
D.	Golgi apparatus	IV.	For storing nutrients

Choose the correct answer from the options given below:
(a) A-III, B-II, C-IV, D-I (b) A-II, B-III, C-I, D-IV
(c) A-III, B-IV, C-II, D-I (d) A-I, B-II, C-III, D-IV

125. Identify the set of correct statements.
A. The flowers of *Vallisneria* are colourful and produce nectar.
B. The flowers of water lily are not pollinated by water.
C. In most of water-pollinated species, the pollen grains are protected from wetting.
D. Pollen grains of some hydrophytes are long and ribbon like.
E. In some hydrophytes, the pollen grains are carried passively inside water.

Choose the correct answer from the options given below.
(a) C, D and E only
(b) A, B, C and D only
(c) A, C, D and E only
(d) B, C, D and E only

126. Match List I with List II.

	List-I		List-II
A.	*Rhizopus*	I.	Mushroom
B.	*Ustilago*	II.	Smut fungus
C.	*Puccinia*	III.	Bread mould
D.	*Agaricus*	IV.	Rust fungus

Choose the correct answer from the options given below.

(a) A-III, B-II, C-IV, D-I
(b) A-I, B-III, C-II, D-IV
(c) A-III, B-II, C-I, D-IV
(d) A-IV, B-III, C-II, D-I

127. *Hind* II always cuts DNA molecules at a particular point called recognition sequence and it consists of
(a) 8 bp (b) 6 bp
(c) 4 bp (d) 10 bp

128. Which of the following is an example of actinomorphic flower?
(a) *Datura* (b) *Cassia*
(c) *Pisum* (d) *Sesbania*

129. The type of conservation in which the threatened species are taken out from their natural habitat and placed in special setting where they can be protected and given special care is called
(a) *in-situ* conservation
(b) biodiversity conservation
(c) semi-conservative method
(d) sustainable development

130. Given below are two statements.

Statement I Chromosomes become gradually visible under light microscope during leptotene stage.

Statement II The beginning of diplotene stage is recognised by dissolution of synaptonemal complex.

In the light of the above statements, choose the correct answer from the options given below.
(a) Both Statement I and Statement II are true
(b) Both Statement I and Statement II are false
(c) Statement I is true but Statement II is false
(d) Statement I is false but Statement II is true

131. Formation of interfascicular cambium from fully developed parenchyma cells is an example for
(a) differentiation
(b) redifferentiation
(c) dedifferentiation
(d) maturation

132. The capacity to generate a whole plant from any cell of the plant is called
(a) totipotency
(b) Micropropagation
(c) differentiation
(d) Somatic hybridisation

133. Match List I with List II.

	List I		List II
A.	Two or more alternative forms of a gene	I.	Back cross
B.	Cross of F_1 progeny with homozygous recessive parent	II.	Ploidy
C.	Cross of F_1 progeny with any of the parents	III.	Allele
D.	Number of chromosome sets in plant	IV.	Test cross

Choose the correct answer from the options given below:
(a) A-I, B-II, C-III, D-IV (b) A-II, B-I, C-III, D-IV
(c) A-III, B-IV, C-I, D-II (d) A-IV, B-III, C-II, D-I

134. The cofactor of the enzyme carboxypeptidase is
(a) zinc (b) niacin (c) flavin (d) haem

135. These are regarded as major causes of biodiversity loss:
A. Overexploitation
B. Co-extinction
C. Mutation
D. Habitat loss and fragmentation
E. Migration
Choose the correct option:
(a) A, C and D only (b) A, B, C and D only
(c) A, B and E only (d) A, B and D only

Section-B

136. Match List I with List II.

	List I (Types of Stamens)		List II (Example)
A.	Monoadelphous	I.	Citrus
B.	Diadelphous	II.	Pea
C.	Polyadelphous	III.	Lily
D.	Epiphyllous	IV.	China rose

Choose the correct answer from the options given below.
(a) A-IV, B-II, C-I, D-III (b) A-IV, B-I, C-II, D-III
(c) A-I, B-II, C-IV, D-III (d) A-III, B-I, C-IV, D-II

137. Match List I with List II.

	List I		List II
A.	GLUT-4	I.	Hormone
B.	Insulin	II.	Enzyme
C.	Trypsin	III.	Intercellular ground substance
D.	Collagen	IV.	Enables glucose transport into cells

Choose the correct answer from the options given below.
(a) A-IV, B-I, C-II, D-III (b) A-I, B-II, C-III, D-IV
(c) A-II, B-III, C-IV, D-I (d) A-III, B-IV, C-I, D-II

138. Identify the step in tricarboxylic acid cycle, which does not involve oxidation of substrate.
(a) Malic acid → Oxaloacetic acid
(b) Succinic acid → Malic acid
(c) Succinyl-Co-A → Succinic acid
(d) Isocitrate → Ñ-ketoglutaric acid

139. Match List I with List II.

	List I		List II
A.	Citric acid cycle	I.	Cytoplasm
B.	Glycolysis	II.	Mitochondrial matrix
C.	Electron transport system	III.	Intermembrane space of mitochondria
D.	Proton gradient	IV.	Inner mitochondrial membrane

Choose the correct answer from the options given below.
(a) A-I, B-II, C-III, D-IV
(b) A-II, B-I, C-IV, D-III
(c) A-III, B-IV, C-I, D-II
(d) A-IV, B-III, C-II, D-I

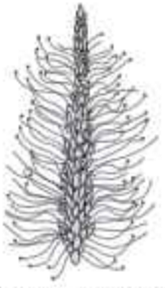
140. Match List I with List II.

	List I		List II
A.	Frederick Griffith	I.	Genetic code
B.	Francois Jacob and Jacque Monod	II.	Semi-conservative mode of DNA replication
C.	Har Gobind Khorana	III.	Transformation
D.	Meselson and Stahl	IV.	*Lac* operon

Choose the correct answer from the options given below.
(a) A-III, B-II, C-I, D-IV
(b) A-III, B-IV, C-I, D-II
(c) A-II, B-III, C-IV, D-I
(d) A-IV, B-I, C-II, D-III

141. Given below are two statements.
Statement I In C_3 plants, some O_2 binds to RuBisCO, hence CO_2 fixation is decreased.
Statement II In C_4 plants, mesophyll cells show very little photorespiration, while bundle sheath cells do not show photorespiration.
In the light of the above statements, choose the correct answer from the options given below.
(a) Both Statement I and Statement II are true
(b) Both Statement I and Statement II are false
(c) Statement I is true but Statement II is false
(d) Statement I is false but Statement II is true

142. Identify the correct description about the given figure.



(a) Wind pollinated plant inflorescence showing flowers with well exposed stamens.
(b) Water pollinated flowers showing stamens with mucilaginous covering.
(c) Cleistogamous flowers showing autogamy.
(d) Compact inflorescence showing complete autogamy

143. Match List I with List II.

	List I		List II
A.	Rose	I.	Twisted aestivation
B.	Pea	II.	Perigynous flower
C.	Cotton	III.	Drupe
D.	Mango	IV.	Marginal placentation

Choose the correct answer from the options given below.
(a) A-II, B-IV, C-I, D-III (b) A-I, B-II, C-III, D-IV
(c) A-IV, B-III, C-II, D-I (d) A-II, B-III, C-IV, D-I

144. Read the following statements and choose the set of correct statements.
In the members of Phaeophyceae,
A. asexual reproduction occurs usually by biflagellate zoospores.
B. sexual reproduction is by oogamous method only.
C. Stored food is in the form of carbohydrates which is either mannitol or laminarin.
D. the major pigments found are chlorophyll-*a*, *c* and carotenoids and xanthophyll.
E. vegetative cells have a cellulosic wall, usually covered on the outside by gelatinous coating of algin.
Choose the correct answer from the options given below.
(a) A, B, C and D only (b) B, C, D and E only
(c) A, C, D and E only (d) A, B, C and E only

145. In an ecosystem if the Net Primary Productivity (NPP) of first trophic level is $100 \times (\text{kcal m}^{-2})\text{yr}^{-1}$.
What will be the GPP (Gross Primary Productivity) of the third trophic level of the same ecosystem
(a) $\frac{x}{10}(\text{kcal m}^{-2})\text{yr}^{-1}$ (b) $x\,(\text{kcal m}^{-2})\text{yr}^{-1}$
(c) $10x\,(\text{kcal m}^{-2})\text{yr}^{-1}$ (d) $\frac{100x}{3x}(\text{kcal m}^{-2})\text{yr}^{-1}$

146. Which of the following statement is correct regarding the process of replication in *E. coli*?
(a) The DNA-dependent DNA polymerase catalyses polymerisation in one direction that is 3′→ 5′
(b) The DNA-dependent RNA polymerase catalyses polymerization in one direction, that is 5′→ 3′
(c) The DNA-dependent DNA polymerase catalyses polymerisation in 5′→ 3′ as well as 3′→ 5′ direction
(d) The DNA-dependent DNA polymerase catalyses polymerisation in 5′→ 3′ direction

147. Which of the following are fused in somatic hybridisation involving two varieties of plants?
(a) Callus (b) Somatic embryos
(c) Protoplasts (d) Pollens

148. Spraying sugarcane crop with which of the following plant growth regulators, increases the length of stem, thus increasing the yield?
(a) Auxin (b) Gibberellin
(c) Cytokinin (d) Abscisic acid

149. Match List I with List II.

	List I		List II
A.	Robert May	I.	Species-area relationship
B.	Alexander von Humboldt	II.	Long term ecosystem experiment using outdoor plots
C.	Paul Ehrlich	III.	Global species diversity at about 7 million
D.	David Tilman	IV.	Rivet popper hypothesis

Choose the correct answer from the options given below.
(a) A-II, B-III, C-I, D-IV (b) A-III, B-I, C-IV, D-II
(c) A-I, B-III, C-II, D-IV (d) A-III, B-IV, C-II, D-I

150. The DNA present in chloroplast is
(a) linear, double-stranded
(b) circular, double-stranded
(c) linear, single-stranded
(d) circular, single-stranded

ZOOLOGY

Section-A

151. Match List I with List II.

	List I		List II
A.	Common cold	I.	*Plasmodium*
B.	Haemozoin	II.	Typhoid
C.	Widal test	III.	Rhinoviruses
D.	Allergy	IV.	Dust mites

Choose the correct answer from the options given below.
(a) A-II, B-IV, C-III, D-I (b) A-I, B-III, C-II, D-IV
(c) A-III, B-I, C-II, D-IV (d) A-IV, B-II, C-III, D-I

152. Match List I with List II.

	List I		List II
A.	Cocaine	I.	Effective sedative in surgery
B.	Heroin	II.	*Cannabis sativa*
C.	Morphine	III.	*Erythroxylum*
D.	Marijuana	IV.	*Papaver somniferum*

Choose the correct answer from the options given below:
(a) A-IV, B-III, C-I, D-II
(b) A-I, B-III, C-II, D-IV
(c) A-II, B-I, C-III, D-IV
(d) A-III, B-IV, C-I, D-II

153. Match List I with List II.

	List I		List II
A.	Fibrous joints	I.	Adjacent vertebrae, limited movement
B.	Cartilaginous joints	II.	Humerus and pectoral girdle, rotational movement
C.	Hinge joints	III.	Skull, do not allow any movement
D.	Ball and socket joints	IV.	Knee, help in locomotion

Choose the correct answer from the options given below.
(a) A-IV, B-II, C-III, D-I
(b) A-I, B-III, C-II, D-IV
(c) A-II, B-III, C-I, D-IV
(d) A-III, B-I, C-IV, D-II

154. Which of the following are autoimmune disorders?
A. Myasthenia gravis
B. Rheumatoid arthritis
C. Gout
D. Muscular dystrophy
E. Systemic Lupus Erythematosus (SLE)
Choose the most appropriate answer from the options given below.
(a) A, B and D only (b) A, B and E only
(c) B, C and E only (d) C, D and E only

155. Which of the following is not a component of Fallopian tube?
(a) Uterine fundus (b) Isthmus
(c) Infundibulum (d) Ampulla

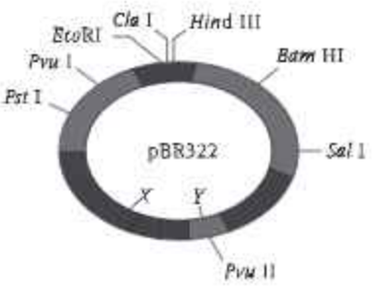
156. The flippers of the penguins and dolphins are the example of the
(a) adaptive radiation (b) natural selection
(c) convergent evolution (d) divergent evolution

157. Match List I with List II.

	List I		List II
A.	α – 1 antitrypsin	I.	Cotton bollworm
B.	*cry* IAb	II.	ADA deficiency
C.	*cry* IAc	III.	Emphysema
D.	Enzyme replacement therapy	IV.	Corn borer

Choose the correct answer form the options given below.
(a) A-II, B-I, C-IV, D-III (b) A-III, B-I, C-II, D-IV
(c) A-III, B-IV, C-I, D-II (d) A-II, B-IV, C-I, D-III

158. The following diagram showing restriction sites in *E. coli* cloning vector pBR322. Find the role of '*X*' and '*Y*' genes.



(a) The gene 'X' is responsible for resistance to antibiotics and 'Y' for protein involved in the replication of plasmid.
(b) The gene 'X' is responsible for controlling the copy number of the linked DNA and 'Y' for protein involved in the replication of plasmid.
(c) The gene 'X' is for protein involved in replication of plasmid and 'Y' for resistance to antibiotics.
(d) Gene 'X' is responsible for recognitions sites and 'Y' is responsible for antibiotic resistance.

159. Given below are two statements; one is labelled as Assertion (A) and the other is labelled as Reason (R).
Assertion (A) Breast-feeding during initial period of infant growth is recommended by doctors for bringing a healthy baby.
Reason (R) Colostrum contains several antibodies absolutely essential to develop resistance for the new born baby.
In the light of the above statements, choose the most appropriate answer from the options given below.
(a) Both A and R are correct and R is the correct explanation of A
(b) Both A and R are correct, but R is not the correct explanation of A
(c) A is correct but R is not correct
(d) A is not correct but R is correct

160. The Ti-plasmid of *Agrobacterium tumefaciens* stands for
(a) Tumour inhibiting plasmid
(b) Tumour independent plasmid
(c) Tumour inducing plasmid
(d) Temperature independent plasmid

161. Match List I with List II.

	List I		List II
A.	*Pleurobrachia*	I.	Mollusca
B.	Radula	II.	Ctenophora
C.	Stomochord	III.	Osteichthyes
D.	Air bladder	IV.	Hemichordata

Choose the correct answer from the options given below.
(a) A-IV, B-II, C-III, D-I (b) A-II, B-I, C-IV, D-III
(c) A-II, B-IV, C-I, D-III (d) A-IV, B-III, C-II, D-I

162. Given below are some stages of human evolution. Arrange them in correct sequence. (Past to Recent)
A. *Homo habilis* B. *Homo sapiens*
C. *Homo neanderthalensis* D. *Homo erectus*
Choose the correct sequence of human evolution from the options given below.
(a) D-A-C-B (b) B-A-D-C
(c) C-B-D-A (d) A-D-C-B

163. Which of the following is not a steroid hormone?
(a) Cortisol (b) Testosterone
(c) Progesterone (d) Glucagon

164. In both sexes of cockroach, a pair of jointed filamentous structures called anal cerci are present on
(a) 5th segment (b) 10th segment
(c) 8th and 9th segment (d) 11th segment

165. Which one of the following factors will not affect the Hardy-Weinberg equilibrium?
(a) Genetic recombination (b) Genetic drift
(c) Gene migration (d) Constant gene pool

166. Match List I with List II.

	List I		List II
A.	Pons	I.	Provides additional space for neurons, regulates posture and balance.
B.	Hypothalamus	II.	Controls respiration and gastric secretions.
C.	Medulla	III.	Connects different regions of the brain.
D.	Cerebellum	IV.	Neuro secretory cells

Choose the correct answer from the options given below.
(a) A-II, B-III, C-I, D-IV
(b) A-III, B-IV, C-II, D-I
(c) A-I, B-III, C-II, D-IV
(d) A-II, B-I, C-III, D-IV

167. Match List I with List II.

	List I		List II
A.	Down's syndrome	I.	11th chromosome
B.	α–thalassemia	II.	X-chromosome
C.	β –thalassemia	III.	21st chromosome
D.	Klinefelter's syndrome	IV.	16th chromosome

Choose the correct answer from the options given below.
(a) A-I, B-II, C-III, D-IV (b) A-II, B-III, C-IV, D-I
(c) A-III, B-IV, C-I, D-II (d) A-IV, B-I, C-II, D-III

168. Which one is the correct product of DNA dependent RNA polymerase to the given template?
3'TACATGGCAAATATCCATTCA5'
(a) 5'AUGUACCGUUUAUAGGUAAGU3'
(b) 5'AUGUAAAGUUUAUAGGUAAGU3'
(c) 5'AUGUACCGUUUAAGGGAAGU3'
(d) 5'ATGTACCGTTTATAGGTAAGT3'

169. Given below are two statements; one is labelled as Assertion A and the other is labelled as Reason (R).
Assertion (A) FSH acts upon ovarian follicles in female and Leydig cells in male.
Reason (R) Growing ovarian follicles secrete oestrogen in female while interstitial cells secrete androgen in male human being.
In the light of the above statements, choose the correct answer from the options given below .
(a) Both A and R are true and R is the correct explanation of A
(b) Both A and R are true, but R is not the correct explanation of A
(c) A is true, but R is false
(d) A is false, but R is true

170. Which of the following is not a natural/traditional contraceptive method?
(a) Coitus interruptus (b) Periodic abstinence
(c) Lactational amenorrhea (d) Vaults

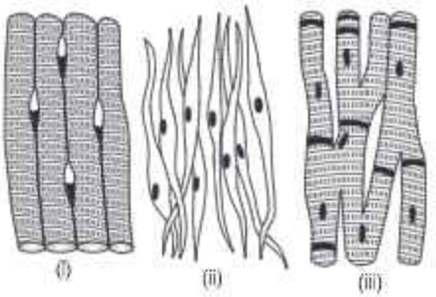
171. Match List I with List II.

	List I		List II
A.	Non-medicated IUD	I.	Multiload 375
B.	Copper releasing IUD	II.	Progestogens
C.	Hormone releasing IUD	III.	Lippes loop
D.	Implants	IV.	LNG-20

Choose the correct answer from the option given below:
(a) A-III, B-I, C-II, D-IV
(b) A-I, B-III, C-IV, D-II
(c) A-IV, B-I, C-II, D-III
(d) A-III, B-I, C-IV, D-II

172. Consider the following statements.
A. Annelids are true coelomates
B. Poriferans are pseudocoelomates
C. Aschelminthes are acoelomates
D. Platyhelminthes are pseudocoelomates
Choose the correct answer from the options given below.
(a) B only (b) A only
(c) C only (d) D only

173. Three types of muscles are given as (i), (ii) and (iii). Identify the correct matching pair along with their location in human body:



Name of muscle/location
(a) (i) Smooth - Toes
(ii) Skeletal - Legs
(iii) Cardiac - Heart
(b) (i) Skeletal - Triceps
(ii) Smooth - Stomach
(iii) Cardiac - Heart
(c) (i) Skeletal - Biceps
(ii) Involuntary-Intestine
(iii) Smooth - Heart
(d) (i) Involuntary - Nose tip
(ii) Skeletal - Bone
(iii) Cardiac - Heart

174. Following are the stages of pathway for conduction of an action potential through the heart.
A. AV bundle B. Purkinje fibres
C. AV node D. Bundle branches
E. SA node
Choose the correct sequence of pathway from the options given below.
(a) E-C-A-D-B (b) A-E-C-B-D
(c) B-D-E-C-A (d) E-A-D-B-C

175. Match List I with List II.

	List I		List II
A.	Lipase	I.	Peptide bond
B.	Nuclease	II.	Ester bond
C.	Protease	III.	Glycosidic bond
D.	Amylase	IV.	Phosphodiester bond

Choose the correct answer from the options given below.
(a) A-IV, B-II, C-III, D-I (b) A-III, B-II, C-I, D-IV
(c) A-II, B-IV, C-I, D-III (d) A-IV, B-I, C-III, D-II

176. Match List I with List II.

	List I		List II
A.	Axoneme	I.	Centriole
B.	Cartwheel pattern	II.	Cilia and flagella
C.	Cristae	III.	Chromosome
D.	Satellite	IV.	Mitochondria

Choose the correct answer from the options given below.
(a) A-IV, B-III, C-II, D-I
(b) A-IV, B-II, C-III, D-I
(c) A-II, B-IV, C-I, D-III
(d) A-II, B-I, C-IV, D-III

177. Match List I with List II.

	List I (Sub-phases of Prophase-I)		List II (Specific Characters)
A.	Diakinesis	I.	Synaptonemal complex formation
B.	Pachytene	II.	Completion of terminalisation of chiasmata
C.	Zygotene	III.	Chromosomes look like thin threads
D.	Leptotene	IV.	Appearance of recombination nodules

Choose the correct answer from the options given below
(a) A-IV, B-II, C-III, D-I
(b) A-I, B-II, C-IV, D-III
(c) A-II, B-IV, C-I, D-III
(d) A-IV, B-III, C-II, D-I

178. Which of the following factors are favourable for the formation of oxyhaemoglobin in alveoli?
(a) High pO_2 and high pCO_2
(b) High pO_2 and lesser H^+ concentration
(c) Low pCO_2 and high H^+ concentration
(d) Low pCO_2 and High temperature

179. Match List I with List II.

	List I		List II
A.	*Pterophyllum*	I.	Hag fish
B.	*Myxine*	II.	Saw fish
C.	*Pristis*	III.	Angel fish
D.	*Exocoetus*	IV.	Flying fish

Choose the correct answer from the options given below.
(a) A-II, B-I, C-III, D-IV
(b) A-III, B-I, C-II, D-IV
(c) A-IV, B-I, C-II, D-III
(d) A-III, B-II, C-I, D-IV

180. Match List I with List II.

	List I		List II
A.	Typhoid	I.	Fungus
B.	Leishmaniasis	II.	Nematode
C.	Ringworm	III.	Protozoa
D.	Filariasis	IV.	Bacteria

Choose the correct answer from the options given below.
(a) A-I, B-III, C-II, D-IV
(b) A-IV, B-III, C-I, D-II
(c) A-III, B-I, C-IV, D-II
(d) A-II, B-IV, C-III, D-I

181. Which of the following statements is incorrect?
(a) A bio-reactor provides optimal growth conditions for achieving the desired product.
(b) Most commonly used bio-reactors are of stirring type.
(c) Bio-reactors are used to produce small scale bacterial cultures.
(d) Bio-reactors have an agitator system, an oxygen delivery system and foam control system.

182. Given below are two statements
Statement I In the nephron, the descending limb of loop of Henle is impermeable to water and permeable to electrolytes.
Statement II The proximal convoluted tubule is lined by simple columnar brush border epithelium and increases the surface area for reabsorption.
In the light of the above statements, choose the correct answer from the option given below.
(a) Both Statement I and Statement II are true
(b) Both Statement I and Statement II are false
(c) Statement I is true but Statement II is false
(d) Statement I is false but Statement II is true

183. Given below are two statements.
Statement I The presence or absence of hymen is not a reliable indicator of virginity.
Statement II The hymen is torn during the first coitus only.
In the light of the above statements, choose the correct answer from the options given below.
(a) Both Statement I and Statement II are true
(b) Both Statement I and Statement II are false
(c) Statement I is true, but Statement II is false
(d) Statement I is false, but Statement II is true

184. Match List I with List II.

	List I		List II
A.	Expiratory capacity	I.	Expiratory reserve volume + Tidal volume + Inspiratory reserve volume
B.	Functional residual capacity	II.	Tidal volume + Expiratory reserve volume
C.	Vital capacity	III.	Tidal volume + Inspiratory reserve volume
D.	Inspiratory capacity	IV.	Expiratory reserve volume + Residual volume

Choose the correct answer from the options given below.
(a) A-II, B-IV, C-I, D-III
(b) A-III, B-II, C-IV, D-I
(c) A-II, B-I, C-IV, D-III
(d) A-I, B-III, C-II, D-IV

185. Following are the stages of cell division
A. Gap 2 phase
B. Cytokinesis
C. Synthesis phase
D. Karyokinesis
E. Gap 1 phase
Choose the correct sequence of stages from the options given below.
(a) C-E-D-A-B
(b) E-B-D-A-C
(c) B-D-E-A-C
(d) E-C-A-D-B

Section-B

186. Given below are two statements.
Statement I Mitochondria and chloroplasts both are double membranes bound organelles.
Statement II Inner membrane of mitochondria is relatively less permeable, as compared to chloroplast.
In the light of the above statements, choose the most appropriate answer from the options given below.
(a) Both Statement I and Statement II are true
(b) Both Statement I and Statement II are false
(c) Statement I is true, but Statement II is false
(d) Statement I is false, but Statement II is true

187. Match List I with List II.

	List I		List II
A.	Mesozoic Era	I.	Lower invertebrates
B.	Proterozoic Era	II.	Fish and Amphibia
C.	Cenozoic Era	III.	Birds and reptiles
D.	Paleozoic Era	IV.	Mammals

Choose the correct answer from the options given below.
(a) A-II, B-I, C-III, D-IV
(b) A-III, B-I, C-II, D-IV
(c) A-I, B-II, C-IV, D-III
(d) A-III, B-I, C-IV, D-II

188. Given below are two statements.
Statement I Gause's competitive exclusion principle states that two closely related species competing for different resources cannot exist indefinitely.
Statement II According to Gause's principle, during competition, the inferior will be eliminated. This may be true if resources are limiting.
In the light of the above statements, choose the correct answer from the options given below.
(a) Both Statement I and Statement II are true.
(b) Both Statement I and Statement II are false.
(c) Statement I is true but Statement II is false.
(d) Statement I is false but Statement II is true.

189. Match List I with List II.

	List I		List II
A.	Unicellular glandular epithelium	I.	Salivary glands
B.	Compound epithelium	II.	Pancreas
C.	Multicellular glandular epithelium	III.	Goblet cells of alimentary canal
D.	Endocrine glandular epithelium	IV.	Moist surface of buccal cavity

Choose the correct answer from the options given below.
(a) A-II, B-I, C-III, D-IV
(b) A-IV, B-III, C-I, D-II
(c) A-III, B-IV, C-I, D-II
(d) A-II, B-I, C-IV, D-III

190. Match List I with List II related to digestive system of cockroach.

	List I		List II
A.	The structures used for storing of food	I.	Gizzard
B.	Ring of 6-8 blind tubules at junction of foregut and midgut.	II.	Gastric caeca
C.	Ring of 100-150 yellow coloured thin filaments at junction of midgut and hindgut.	III.	Malpighian tubules
D	The structures used for grinding the food.	IV.	Crop

Choose the correct answer from the options given below.
(a) A-IV, B-II, C-III, D-I (b) A-I, B-II, C-III, D-IV
(c) A-IV, B-III, C-II, D-I (d) A-III, B-II, C-IV, D-I

191. Choose the correct statement given below regarding juxta medullary nephron.
(a) Juxta medullary nephrons are located in the columns of Bertini.
(b) Renal corpuscle of juxta medullary nephron lies in the outer portion of the renal medulla.
(c) Loop of Henle of juxta medullary nephron runs deep into medulla.
(d) Juxta medullary nephrons outnumber the cortical nephrons.

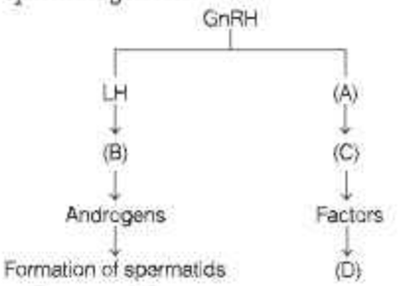
192. Match List I with List II.

	List I		List II
A.	RNA polymerase	I.	*sn*RNPs
B.	Termination of transcription	II.	Promotor
C.	Splicing of exons	III.	*rho* factor
D.	TATA box	IV.	*sn*RNAs, *t*RNA

Choose the correct answer from the options given below :
(a) A-II, B-IV, C-I, D-III
(b) A-III, B-II, C-IV, D-I
(c) A-III, B-IV, C-I, D-II
(d) A-IV, B-III, C-I, D-II

193. Given below are two statements.
Statement I The cerebral hemispheres are connected by nerve tract known as corpus callosum.
Statement II The brain stem consists of the medulla oblongata, pons and cerebrum.
In the light of the above statements, choose the most appropriate answer from the options given below.
(a) Both Statement I and Statement II are true
(b) Both Statement I and Statement II are false
(c) Statement I is true, but Statement II is false
(d) Statement I is false, but Statement II is true

194. Identify the correct option (A), (B), (C), (D) with respect to spermatogenesis.



Formation of spermatids
(a) FSH, Leydig cells, Sertoli cells, Spermiogenesis
(b) ICSH, Interstitial cells, Leydig cells, spermiogenesis
(c) FSH, Sertoli cells, Leydig cells, Spermatogenesis
(d) ICSH, Leydig cells, Sertoli cells, Spermatogenesis

195. As per ABO blood grouping system, the blood group of father is B^+, mother is A^+ and child is O^+. Their respective genotype can be
A. $I^Bi / I^Ai / ii$ B. $I^BI^B / I^AI^A / ii$
C. $ii^B / ii^A / I^AI^B$ D. $I^Ai / I^Bi / I^Ai$
E. $ii^B / ii^A / I^AI^B$
Choose the most appropriate answer from the options given below.
(a) A only (b) B only
(c) C and B only (d) D and E only

196. Given below are two statements :
Statement I Bone marrow is the main lymphoid organ where all blood cells including lymphocytes are produced.
Statement II Both bone marrow and thymus provide micro environments for the development and maturation of T-lymphocytes.
In the light of above statements, choose the most appropriate answer from the options given below
(a) Both Statement I and Statement II are true
(b) Both Statement I and Statement II are false
(c) Statement I is true, but Statement II is false
(d) Statement I is false, but Statement II is true

197. Regarding catalytic cycle of an enzyme action, select the correct sequential steps.
A. Substrate enzyme complex formation.
B. Free enzyme ready to bind with another substrate.
C. Release of products.
D. Chemical bonds of the substrate broken.
E. Substrate binding to active site.
Choose the correct answer from the options given below.
(a) E, A, D, C, B (b) A, E, B, D, C
(c) B, A, C, D, E (d) E, D, C, B, A

198. Match List I with List II.

	List I		List II
A.	P-wave	I.	Heart muscles are electrically silent.
B.	QRS-complex	II.	Depolarisation of ventricles.
C.	T-wave	III.	Depolarisation of atria.
D.	T-P gap	IV.	Repolarisation of ventricles.

Choose the correct answer from the options given below.
(a) A-I, B-III, C-IV, D-II (b) A-III, B-II, C-IV, D-I
(c) A-II, B-III, C-I, D-IV (d) A-IV, B-II, C-I, D-III

199. Match List I with List II.

	List I		List II
A.	Exophthalmic goitre	I.	Excess secretion of cortisol, moon face and hyperglycemia.
B.	Acromegaly	II.	Hyposecretion of thyroid hormone and stunted growth.
C.	Cushing's syndrome	III.	Hypersecretion of thyroid hormone and protruding eye balls.
D.	Cretinism	IV.	Excessive secretion of growth hormone.

Choose the correct answer from the options given below.
(a) A-I, B-III, C-II, D-IV
(b) A-IV, B-II, C-I, D-III
(c) A-III, B-IV, C-II, D-I
(d) A-III, B-IV, C-I, D-II

200. The following are the statements about non-chordates.
A. Pharynx is perforated by gill slits.
B. Notochord is absent.
C. Central nervous system is dorsal.
D. Heart is dorsal if present.
E. Post anal tail is absent.
Choose the most appropriate answer from the options given below.
(a) A and C only (b) A, B and D only
(c) B, D and E only (d) B, C and D only

Answers

Physics

1. (b)	2. (d)	3. (c)	4. (a)	5. (c)	6. (a)	7. (d)	8. (b)	9. (b)	10. (c)
11. (b)	12. (c)	13. (c)	14. (b)	15. (b)	16. (c)	17. (d)	18. (b)	19. (d)	20. (a)
21. (c)	22. (a)	23. (a)	24. (b)	25. (b)	26. (d)	27. (a)	28. (d)	29. (a)	30. (a)
31. (a)	32. (b)	33. (d)	34. (a)	35. (d)	36. (c)	37. (b)	38. (b)	39. (d)	40. (b)
41. (a)	42. (b)	43. (d)	44. (b)	45. (a)	46. (b)	47. (b)	48. (b)	49. (b)	50. (b)

Chemistry

51. (a)	52. (d)	53. (a)	54. (a)	55. (a)	56. (b)	57. (d)	58. (d)	59. (a)	60. (b)
61. (a)	62. (a)	63. (b)	64. (c)	65. (d)	66. (a)	67. (a)	68. (c)	69. (a)	70. (d)
71. (d)	72. (c)	73. (d)	74. (d)	75. (a)	76. (a)	77. (c)	78. (b)	79. (a)	80. (c)
81. (c)	82. (a)	83. (b)	84. (b)	85. (c)	86. (a)	87. (d)	88. (b)	89. (a)	90. (a)
91. (d)	92. (a)	93. (b)	94. (a)	95. (d)	96. (b)	97. (b)	98. (d)	99. (a)	100. (a)

Botany

101. (b)	102. (c)	103. (b)	104. (a)	105. (a)	106. (c)	107. (c)	108. (d)	109. (c)	110. (c)
111. (d)	112. (c)	113. (b)	114. (b)	115. (c)	116. (b)	117. (d)	118. (d)	119. (d)	120. (a)
121. (c)	122. (c)	123. (b)	124. (a)	125. (d)	126. (a)	127. (b)	128. (a)	129. (b)	130. (a)
131. (c)	132. (a)	133. (c)	134. (a)	135. (d)	136. (a)	137. (a)	138. (c)	139. (b)	140. (b)
141. (c)	142. (a)	143. (a)	144. (c)	145. (c)	146. (d)	147. (c)	148. (b)	149. (b)	150. (b)

Zoology

151. (c)	152. (d)	153. (d)	154. (b)	155. (a)	156. (c)	157. (c)	158. (b)	159. (a)	160. (c)
161. (b)	162. (d)	163. (d)	164. (b)	165. (d)	166. (b)	167. (c)	168. (a)	169. (d)	170. (d)
171. (d)	172. (b)	173. (b)	174. (a)	175. (c)	176. (d)	177. (c)	178. (b)	179. (b)	180. (b)
181. (c)	182. (b)	183. (c)	184. (a)	185. (d)	186. (c)	187. (d)	188. (d)	189. (c)	190. (a)
191. (c)	192. (d)	193. (c)	194. (a)	195. (a)	196. (a)	197. (a)	198. (b)	199. (d)	200. (c)

Answer with Explanations

Physics

1. *(b)* The tension in the string will be equal to centripetal force is,

$$F_c = \frac{mv^2}{r} = m\omega^2 r$$

According to question, $T = m\omega^2 r$

and $T' = m(2\omega)^2 r$

$$\frac{T'}{T} = \frac{m(4\omega^2)r}{m\omega^2 r} \Rightarrow T' = 4T$$

2. *(d)* The direction of motion changes continuously in a circular motion, therefore circular motion is an accelerated motion.
So, uniform circular motion have varying velocity as well as varying acceleration.

3. *(c)* Output from the truth table

$$Y = \overline{A}B + A\overline{B} = \overline{B}(A + \overline{A})$$

$$Y = \overline{B}$$

So, correct option is (c).

4. *(a)* The direction of induced current is always such that it opposes the cause which created it. For solenoid-1, the magnet is moving away. Thus, near the *B*-point, *S*-pole will be generated.



Thus, the current in solenoid-1 will be from *A* to *B*.
Similarly, the current in the solenoid-2 will be from *D* to *C*.

5. *(c)* We know that, potential at point due to an ideal point at any point on axial line is given

$$V = \pm \frac{1}{4\pi\varepsilon_0}\frac{p}{r^2}$$

Putting the values of p and r, we get

$$V = \pm 9\times10^9 \times \frac{4\times10^{-6}}{(2)^2} = \pm 9\times10^3 \text{ V}$$

Hence, A is true but R is false.

6. *(a)* For a diamagnetic material magnetic susceptibility χ is small and negative, is, $0 > \chi \geq -1$.
For a ferromagnetic material, $\chi >> 1$.
For a paramagnetic material, $0 < \chi < \varepsilon$ (ε is a small positive number) and for non-magnetic material, $\chi = 0$.
Hence, correct option is (a).

7. *(d)* Given, magnetic field $B = 0.049$ T

Time period, $T = \frac{5}{20} = \frac{1}{4}$ s

Moment of inertia, $I = 9.8\times10^{-6}$ kg m^2

The time period of oscillation of a magnetic needle in a magnetic field is given by $T = 2\pi\sqrt{\frac{I}{MB}}$

where, M is the magnetic moment of the needle.

Thus, $T = 2\pi\sqrt{\frac{9.8\times10^{-6}}{M\times0.049}}$

$$\Rightarrow \left(\frac{1}{4}\right)^2 = (2\pi)^2\left(\frac{9.8\times10^{-6}}{M\times4.9\times10^{-2}}\right)$$

$$\Rightarrow \frac{1}{16} = \frac{4\pi^2\times2\times10^{-4}}{M}$$

$$\Rightarrow M = 16\times8\times\pi^2\times10^{-4}$$
$$= 128\pi^2\times10^{-4}\text{ Am}^2$$
$$= 1280\pi^2\times10^{-5}\text{ Am}^2$$

8. *(b)* We know that, transformation ratio,

$$r = \frac{N_s}{N_p} = \frac{V_s}{V_p}$$

$$\because \quad \frac{N_p}{N_s} = \frac{1}{2} = \frac{V_p}{V_s}$$

$$\Rightarrow V_s : V_p = 2:1$$

9. *(b)* Given, 1 MSD = 0.1 mm

$$(N+1)\text{ VSD} = N\text{ MSD}$$

$$\therefore \quad 1\text{ VSD} = \left(\frac{N}{N+1}\right)\text{MSD}$$

Least count or vernier constant,

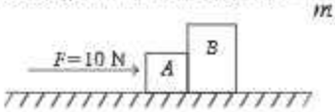
$$1\text{MSD} - 1\text{VSD} = \left[1 - \frac{N}{N+1}\right]\text{MSD}$$

$$= \left(\frac{1}{N+1}\right)\text{MSD} = \left(\frac{0.1}{N+1}\right)\text{mm}$$

$$= \frac{1}{100(N+1)}\text{cm}$$

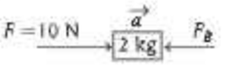
10. *(c)* Net force on *A* and *B* = 10 N

Acceleration of both *A* and *B* $= \frac{F}{m}$



$$a = \frac{10}{5} = 2\text{ ms}^{-2}$$

For block *A*,

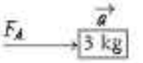


F_B = force exerted by block *B* on block *A*.

$$F - F_B = 2\times a$$

$$\Rightarrow \quad 10 - F_B = 2\times2 \Rightarrow F_B = 6\text{N}$$

For block *B*,



Force exerted by *A* on B,

$$F_A = 3a = 6\text{ N}$$

11. *(b)* Given, $x = 5\sin\left(\pi t + \frac{\pi}{3}\right)$ m ...(i)

The standard Eq. for SHM is

$$x = A\sin(\omega t + \phi) \quad \ldots(ii)$$

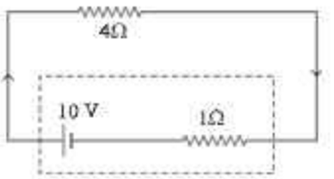
Thus, comparing Eqs. (i) and (ii), we get

$$A = 5\text{ m and } \omega = \pi$$

$$\Rightarrow \quad \frac{2\pi}{T} = \pi \Rightarrow T = 2s$$

Hence, option (b) is correct.

12. *(c)* Given,
EMF, $\varepsilon = 10$ V
Internal resistance, $r = 1\,\Omega$
External resistance, $R = 4\,\Omega$



Terminal voltage, $V = \varepsilon - Ir$

and $I = \frac{\varepsilon}{R+r} = \frac{10}{4+1} = 2\text{ A}$

Thus, $V = 10 - 2\times1 = 8$ V

13. *(c)* Atoms are electrically neutral as they contain equal number of protons and electrons. Statement II is incorrect because atom of radioactive elements are not stable.
So, option (c) is correct.

14. *(b)* Photon is quanta of energy. It is a massless and chargeless particle. The rest mass of photon is zero. In free space, it moves with velocity c and energy of photon is equal to $h\nu$. The momentum of the photon is given as

$$p = \frac{E}{c} = \frac{h\nu}{c}$$

15. *(b)* The wavelength of spectral line is given by

$$\frac{1}{\lambda} = R\left[\frac{1}{n_1^2} - \frac{1}{n_2^2}\right]$$

A. $\frac{1}{\lambda} = 1.09737\times10^{7}\left[\frac{1}{2^2} - \frac{1}{3^2}\right]$

$= 1.09737\times10^{7}\left[\frac{1}{4} - \frac{1}{9}\right]$

$= 1.09737\times10^{7}\left[\frac{5}{36}\right]$

$\Rightarrow \lambda = \frac{36\times10^{-7}}{5\times1.09737} = 6.56114\times10^{-7}$

$\simeq 656.3$ nm

B. $\lambda = \frac{10^{-7}}{1.09737\left[\frac{1}{2^2} - \frac{1}{4^2}\right]}$

$= \frac{10^{-7}\times16}{1.09737\times3} = 4.8601\times10^{-7}$ m

$\simeq 486.1$ nm

C. $\lambda = \frac{10^{-7}}{1.09737\times\left[\frac{1}{2^2} - \frac{1}{5^2}\right]}$

$= \frac{10^{-7}\times100}{1.09737\times21} = 4.3393\times10^{-7}$ m

$\simeq 434.1$ nm

D. $\lambda = \frac{10^{-7}}{1.09737\times\left[\frac{1}{2^2} - \frac{1}{6^2}\right]}$

$= \frac{10^{-7}\times144}{1.09737\times32} \simeq 410.2$ nm

Hence, A - III, B - IV, C - II, D - 1

16. *(c)* Given, number of turns $N = 100$

radius, $r = 10\text{ cm} = 10\times10^{-2}$ m

current, $i = 7$A

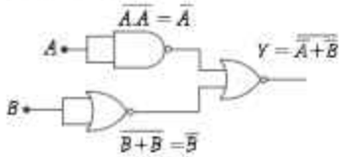
Magnetic field at the centre of a solenoid,

$$B = \frac{\mu_0 NI}{2r} = \frac{4\pi\times10^{-7}\times100\times7}{2\times10\times10^{-2}}$$

$= 44\times10^{-4} = 4.4\times10^{-3}$ T

$= 4.4$ mT

17. *(d)* The given combination of gates can be taken as



$Y = \overline{\overline{A} + \overline{B}} = \overline{\overline{A}}\cdot\overline{\overline{B}} = A\cdot B$

Hence, the output Y of the given logic gate is similar to AND gate.

18. *(b)* Given,

wire of length l is divided into 10 parts each having $\frac{100}{10}\Omega = 10\Omega$ resistance.

According to question, 5 parts are connected in series

i.e., $R_1 = 5\times10 = 50\Omega$

5 parts are connected in parallel

i.e., $R_2 = \frac{R}{5} = \frac{10}{5} = 2\Omega$

Then, both R_1 and R_2 are connected in series

Thus, $R_{net} = R_1 + R_2$

$= 50 + 2 = 52\Omega$

19. *(d)* Given,

$$^{290}_{82}X \xrightarrow{\alpha} Y \xrightarrow{e^+} Z \xrightarrow{\beta^-} P \xrightarrow{e^-} Q$$

From $X \to Y$, α-particle is emitted thus $^{286}_{80}Y$.

Now, $Y \to Z$, e^+ is emitted, thus $^{286}_{79}Z$.

Again $Z \to P$, β^- is emitted thus $^{286}_{80}P$.

Again $P \to Q$, e^- is emitted thus $^{286}_{81}Q$.

20. *(a)* Length of wire $= 1$ m

Elastic limit, $E = 8\times10^{8}$ Nm^{-2}

Young's modulus $Y = 2\times10^{11}$ Nm^{-2}

We know that, $Y = \frac{F}{A}\cdot\frac{l}{\Delta l}$

Here, $\frac{F}{A}$ = Elastic limit E

$\Rightarrow Y = E\frac{l}{\Delta l} \Rightarrow \Delta l = \frac{E}{Y}l$

$\Rightarrow \Delta l = \frac{8\times10^{8}}{2\times10^{11}}\times1 = 4\times10^{-3}$ m

$= 4$ mm

21. *(c)* When YDSE is performed in white light instead of monochromatic light, all wavelengths shows their maxima and minima in order of their wavelength. Thus, a central bright is observed along with few fringes that will be different colours.

22. *(a)* Given,

$x = 2t - 1$ unit

Force, $F = 5$ N

Instantaneous power, $P = F.v$

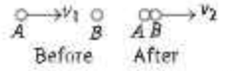
Thus, $P = F\frac{dx}{dt} = 5\times\frac{d(2t-1)}{dt}$

$= 5\times2 = 10$ unit

23. *(a)* A is correct but B is incorrect.

In reverse biased p-n junction, the current is due to minority charge carriers, which is very small and hence measured in terms of μA.

24. *(b)* Let both masses be m



Thus, applying conservation of momentum, we get

$m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2$

Both masses A and B stick together as a system.

So, $mv_1 = 2mv_2$ $[\because u_2 = 0]$

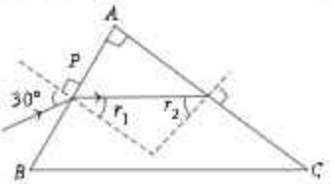
$v_1 = 2v_2 \Rightarrow \frac{v_1}{v_2} = \frac{2}{1}$

Hence, option (b) is correct.

25. *(b)* Given, the rays in the prism is parallel to base.

$A = r_1 + r_2$

i.e., $r_2 = 90 - r_1$ ...(i)



For AB interface,

$\sin 30° = \mu\sin r_1$

$\Rightarrow \sin r_1 = \frac{1}{2\mu}$ ...(ii)

For AC interface,

$\mu\sin r_2 = 1\sin 90° \Rightarrow \sin r_2 = \frac{1}{\mu}$

$\Rightarrow \sin(90 - r_1) = \frac{1}{\mu}$

$\cos r_1 = \frac{1}{\mu}$ ...(iii)

We know that, $\sin^2\theta + \cos^2\theta = 1$

From Eqs. (ii) and (iii), we get

$\sin^2 r_1 + \cos^2 r_1 = 1$

$\Rightarrow \left(\frac{1}{2\mu}\right)^2 + \left(\frac{1}{\mu}\right)^2 = 1 \Rightarrow \frac{1}{4\mu^2} + \frac{1}{\mu^2} = 1$

$\Rightarrow \frac{1}{\mu^2}\left(\frac{1}{4} + \frac{1}{1}\right) = 1$

$\Rightarrow \mu^2 = \frac{5}{4} \Rightarrow \mu = \frac{\sqrt{5}}{2}$

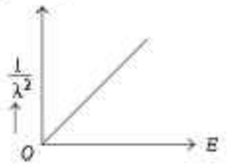
26. *(d)* We know that $\lambda = \frac{h}{p} = \frac{h}{\sqrt{2mE}}$

Here, E = kinetic energy

Thus, $\lambda^2 = \frac{h^2}{2mE} \Rightarrow \frac{1}{\lambda^2} = \frac{2mE}{h^2}$

$\propto E$

Hence, option (d) is correct.



27. *(a)* Strain and angle are dimensionless quantities as solid angle.
Arc has dimension of length.
Stress has dimension of pressure.

28. *(d)* At Brewster's angle, the angle between refracted and reflected rays is 90°. The reflected ray is completely linearly polarised and the refracted rays is partially polarised.

29. *(a)* Given, moment of inertia $I = 2400\ \text{g cm}^2$
mass of rod, $m = 400\ \text{g}$
We know that moment of inertia of a thin rod,

$$I = \frac{ml^2}{12} \Rightarrow \frac{12 \times I}{m} = l^2$$

$$\Rightarrow \quad l^2 = \frac{12 \times 2400}{400} \Rightarrow l^2 = 12 \times 6$$

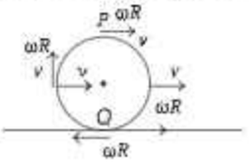
$$\Rightarrow \quad l = \sqrt{72} = 8.48 \simeq 8.5\ \text{cm}$$

30. *(a)* Radius of disc, $r = 4.5\ \text{cm}$
$= 4.5 \times 10^{-2}\ \text{m}$
Surface tension, $S = 0.07\ \text{Nm}^{-1}$
The force required to take away circular disc from the surface of water,

$$F = S \times 2\pi r$$
$$= 0.07 \times 2 \times 3.14 \times 4.5 \times 10^{-2}$$
$$= 1.9782 \times 10^{-2}\ \text{N} = 19.8\ \text{mN}$$

31. *(a)* Work done in an isochoric process is always zero.

32. *(b)* In rolling motion, wheel is rolling with velocity v. All the points move with different velocity as shown



Thus, the velocity of P is greater than velocity at Q.

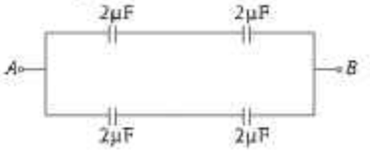
33. *(d)* We know that, $g = \frac{GM}{R^2}$
According to question,

$$g = \frac{G\left(\frac{M}{10}\right)}{\left(\frac{R}{2}\right)^2} = \frac{GM \times 4}{R^2 \times 10} = \frac{GM}{R^2} \times \frac{2}{5}$$

Hence, $\frac{g'}{g} = \frac{\frac{GM}{R^2} \times \frac{2}{5}}{\frac{GM}{R^2}}$

$$\Rightarrow \quad g' = \frac{2}{5} \times g = \frac{2}{5} \times 9.8 = 3.92\ \text{ms}^{-2}$$

34. *(a)* Given combination of capacitors represent Wheatstone bridge, thus central capacitor can be removed.



Capacitance of each branch,

$$C = \frac{2}{2} = 1\mu\,\text{F}$$

Capacitance between A and B,
$C_{net} = 1 + 1 = 2\mu\text{F}$

35. *(d)* Since, there is no electric field inside the shell the potential at any point inside the shell is equal to the potential on the surface of the shell is

$$V = \frac{1}{4\pi\varepsilon_0} \frac{q}{r}$$

$$V_C - V_P = 0$$

36. *(c)* We know that

$$a = \frac{dv}{dt}$$

For v = constant, $a = 0$
Thus, for v - t graph when v is positive and linearly increasing a will be positive constant and if v is decreasing linearly a will be negative constant.
Hence, option (c) is correct.

37. *(b)* Time period of a bob is given by,

$$T = 2\pi\sqrt{\frac{l}{g}}$$

According to question, $T' = 2\pi\sqrt{\frac{l}{2g}}$

$$= \frac{2\pi\sqrt{l}}{\sqrt{2g}}$$

Thus, $\frac{T'}{T} = \frac{T}{\sqrt{2}} = \frac{x}{2}T$

Hence, $x = \sqrt{2}$

38. *(b)* Given, $C = 10\mu\text{F} = 10 \times 10^{-6}\ \text{F}$
Potential, $V = 210\ \text{V}$
Frequency, $f = 50\ \text{Hz}$
Reactance of circuit, $X_C = \frac{1}{\omega C} = \frac{1}{2\pi f C}$

$$= \frac{1}{2 \times 3.14 \times 50 \times 10 \times 10^{-6}} = \frac{10^3}{3.14}$$

Peak current, $I_m = \frac{V\sqrt{2}}{X_C} = \frac{210 \times \sqrt{2}}{\frac{10^3}{3.14}}$

$$= \frac{3.14 \times 210 \times \sqrt{2}}{1000} = 0.93\ \text{A}$$

39. *(d)* Charle's law states that volume of a gas is proportional to its temperature on Kelvin scale at constant pressure.
It is clear for p_1, T is higher.
Thus, $p_1 > p_2 > p_3$.

Alternate Solution

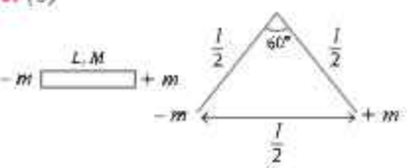
We know that $pV = nRT$

$$\frac{T}{V} = \frac{p}{nR}$$

$$\text{Slope} = \frac{p}{nR}$$

i.e., slope $\propto p$
Hence, $p_1 > p_2 > p_3$

40. *(b)*



The magnetic moment is given by
$M = m \cdot L$
New magnetic moment, $M' = m \cdot L'$

$$= m \cdot \frac{l}{2} = \frac{ml}{2}$$

Thus, $M' = \frac{M}{2}$

41. *(a)* Given altitude, $h = 2R$
From surface, $r_1 = R$
and $r_2 = 2R + R = 3R$
Then, energy required to launch a satellite from orbit r_1 to r_2 is sum of difference in energies in the orbit (ΔE) and energy on the surface of the earth.

$$\Delta E = GMm\left(\frac{1}{r_1} - \frac{1}{r_2}\right) = GMm\left(\frac{1}{R} - \frac{1}{3R}\right)$$
$$= \frac{2}{3}\left(\frac{GMm}{R}\right)$$

Energy on surface $= \frac{1}{2}mv^2$

$$v^2 = \frac{GM}{r_{orbit}} = \frac{GM}{3R}$$

∴ Minimum energy

$$= \Delta E + \frac{GMm}{6R} = \frac{2}{3}\left(\frac{GMm}{R}\right) + \left(\frac{GMm}{6R}\right)$$
$$= \frac{5}{6}\frac{GMm}{R}$$

42. *(b)* When a capacitor is charged through a battery, its opposite plates are charged to a maximum value. There is displacement current as the capacitor is being charged, $I_d = \varepsilon \cdot \frac{d\phi_E}{dt}$

Due to charging, ϕ_E changes and hence displacement current equal to conduction current arises while charging. When capacitor gets fully charged, the displacement current became zero.

43. *(d)* Electromagnetic waves are generated by accelerated charged particles.

Hence, option (d) is incorrect.

44. *(b)* Given,

Young's modulus, $Y = 0.5 \times 10^{11}\ \text{Nm}^{-2}$

Coefficient of liner thermal expansion,

$$\alpha = 10^{-5}{}^{\circ}\text{C}^{-1}$$

Length, $l = 1\ \text{m}$

Area of cross-section, $A = 10^{-3}\text{m}^2$

$$\Delta T = 100^{\circ}\text{C}$$

We know that, stress $= Y \cdot \frac{\Delta l}{l}$

Here, $\Delta l = l\alpha\Delta T = 1 \times 10^{-5} \times 100 = 10^{-3}$

$$\Rightarrow \quad \frac{F}{A} = Y \cdot \frac{10^{-3}}{1}$$

Compressive force,

$$F = 0.5 \times 10^{11} \times 10^{-3} \times 10^{-3}$$

$$= 0.5 \times 10^{5} = 50 \times 10^{3}\ \text{N}$$

45. *(a)* For a diode to be active it must be forward biased. In option (b) and (d), diode is reverse biased.

Hence, option (b) and (d) cannot achieve bridge balance.

For option (a) :

Balance bridge condition is $\frac{10}{15} = \frac{10}{5+r}$

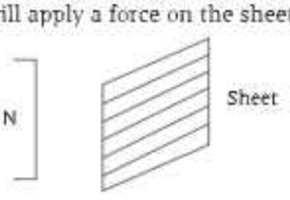
If $r = 10\Omega$, bridge can be balanced.

For option (c) : $\frac{10}{15} = \frac{10}{5 \| r}$

In this case, the resistance of 5Ω branch can only decrease.

∴This circuit can not achieve balance condition.

46. *(b)* If the sheet is magnetic in nature, then the flux will be linked through the sheet and a small amount of force will be needed to held the sheet. The pole will apply a force on the sheet.



Also, to move the sheet towards or away from the magnetic pole, a force will be required.

If sheet is non-magnetic then no force will be required either to hold it or move it when the sheet is non-conducting.

47. *(b)* We know that,

$$C = \frac{\varepsilon_0 A}{d}$$

$$E = \frac{1}{2}CV^2 \text{ and } Q = C.V$$

As the distance between the plates decreases capacitance increases, therefore energy increases and also charge stored is increased.

Hence, A, C, and E are correct.

48. *(b)* Given,

$P_1 = 1\ \text{kW}$

$P_2 = 2\ \text{kW}$

$P_{\text{Series}} = \frac{P_1 P_2}{P_1 + P_2}$ and $P_{\text{Parallel}} = P_1 + P_2$

According to question,

$$\frac{P_{\text{Series}}}{P_{\text{Parallel}}} = \frac{\frac{1\times 2}{1+2}}{3\times 3} = \frac{2}{9}$$

49. *(b)* Given,

Focal length of objective, $f_0 = 140\ \text{cm}$

Focal length of eyepiece, $f_e = 5\ \text{cm}$

Magnifying power, $M = \left[\frac{f_0}{f_e}\right]$

$$= \left[\frac{140}{5}\right] = 28$$

50. *(b)* Given, $F = \alpha t^2 + \beta t$

We know that dimensions on the both sides will be the same, i.e.,

$$[F] = [\alpha t^2] = [\beta t]$$

$\Rightarrow [\alpha][T^2] = [ML^1T^{-2}] \Rightarrow \alpha = [M^1L^1T^{-4}]$

and $[\beta][T] = [M^1LT^{-2}] \Rightarrow \beta = [M^1L^1T^{-3}]$

(a) $\frac{[\beta t]}{[\alpha]} = \frac{[M^1L^1T^{-2}]}{[M^1L^1T^{-4}]} = [T^{+2}]$

(b) $\frac{[\alpha t]}{[\beta]} = \frac{[M^1L^1T^{-4}][T]}{[M^1L^1T^{-3}]} = [M^0L^0T^0]$

Hence, option (b) is correct.

Chemistry

51. *(a)* Spin only magnetic moment is given by, $\mu_s = \sqrt{n(n+2)}$

where, n = number of unpaired electrons

A. $Ti^{3+} \longrightarrow 3d^1$, $n = 1$

$$\mu_s = \sqrt{1(1+2)} = \sqrt{3}$$

B. $Cr^{2+} \longrightarrow 3d^4$, $n = 4$

$$\mu_s = \sqrt{4(4+1)} = \sqrt{20}$$

C. $Mn^{2+} \longrightarrow 3d^5$, $n = 5$

$$\mu_s = \sqrt{5(5+1)} = \sqrt{30}$$

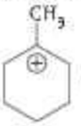
D. $Fe^{2+} \longrightarrow 3d^6$, $n = 4$

$$\mu_s = \sqrt{4(4+1)} = \sqrt{20}$$

E. $Sc^{3+} \longrightarrow 3d^0$, $\mu_s = 0$

Hence, B and D have same value of magnetic moment.

52. *(d)* The stability of carbocation is depending on number of α-hydrogen. The more the number of alpha hydrogen atoms, more is the stability of the carbocation.



has 7 α-hydrogen, which is highest among all given structures.

53. *(a)* Both statement I and statement II are true.

54. *(a)* The correct match is

A-I, B-IV, C-II, D-III.

A. $NH_3 \longrightarrow sp^3$ hybridised with 1 lone pair $\longrightarrow$ trigonal pyramidal.

B. $BrF_5 \longrightarrow sp^3d^2$ hybridised with 1 lone pair $\longrightarrow$ square pyramidal

C. $XeF_4 \longrightarrow sp^3d^2$ hybridised with 2 lone pairs $\longrightarrow$ square planar

D. $SF_6 \longrightarrow sp^3d^2$ hybridised with zero lone pair $\longrightarrow$ octahedral

55. *(a)* In 4 moles of He, number of He atom $= 4 \times N_A$

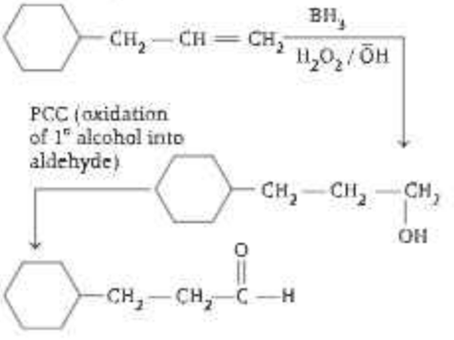
In 4u of He, number of He atom = 1

In 4g of He, number of He atom $= N_A$

In 2.271098 L of He, number of He atom $= \frac{2.27}{22.7} \times N_A = 0.1N_A$

Hence, option (a) is correct.

56. *(b)* The complete reaction is as follows,



57. *(d)* The correct match is A-II, B-III, C-IV, D-I.

58. *(d)* The reactivity order of alcohols toward Lucas' reagent is 3° alcohols > 2° alcohols > 1° alcohol.

(a) $CH_3—CH_2—CH_2—CH_2—OH$
(1° alcohol)

(b) $CH_3—CH_2—\underset{\displaystyle CH_3}{\underset{|}{CH}}—OH$
(2° alcohol)

(c) $CH_3—\underset{\displaystyle CH_3}{\underset{|}{CH}}—CH_2—OH$
(1° alcohol)

(d) $CH_3—\underset{\displaystyle CH_3}{\overset{\displaystyle CH_3}{\overset{|}{\underset{|}{C}}}}—OH$
(3° alcohol)

Hence, option (d) is correct.

59. *(a)* In

$PCl_5(g) \rightleftharpoons PCl_3(g) + Cl_2(g)$

$K_p = K_C(RT)^{\Delta n_g}$

when $\Delta n_g = 0$ then $K_p = K_C$

$\Delta n_g \neq 0$ then $K_p \neq K_C$

In $PCl_5(g)$,

$\Delta n_g \neq 0 \quad \therefore \quad K_p \neq K_C$

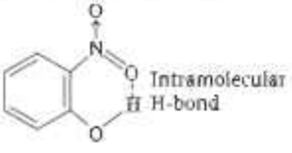
Rest all options has $\Delta n_g = 0$.

So $K_p = K_C$ for them.

60. *(b)* The correct match is A-III, B-IV, C-I, D-II.

61. *(a)* Both statement I and statement II are true.

62. *(a)* Intramolecular hydrogen bonding is present in *o*-nitrophenol.



63. *(b)* The process of changing the state directly from solid to vapour is called as sublimation.

Hence, option (b) is correct.

64. *(c)* The statement given in A, C, D are correct in regard to increase in entropy.

A. When liquid evaporates to vapour, randomness increases, so entropy increases.

C. $\Delta n_g = 2 - 0 = 2$

$\Delta n_g = +ve \longrightarrow \Delta S = +ve$

$\Rightarrow$ Entropy increases.

D. $\Delta n_g = 2 - 1 = 1$

$\Delta n_g = +ve \longrightarrow \Delta S = +ve$

$\Rightarrow$ Entropy increases.

Statement given in B is incorrect, because, when temperature decreases, entropy decreases.

65. *(d)* Among the given elements, only Po does not show –2 oxidation state due to its highly metallic nature.

66. *(a)* The correct match is

A-II, B-IV, C-I, D-III.

A. $2H_2O \longrightarrow O_2 + 4H^+ + 4e^-$

2 mol H_2O require 4F charge

1 mol H_2O will require 2F charge.

B. $8H^+ + MnO_4^- + 5e^- \longrightarrow Mn^{2+} + 4H_2O$

1 mole of MnO_4^- will require 5F charge.

C. $CaCl_2 + 2e^- \longrightarrow Ca + 2Cl^-$

1 mole of Ca will require 2F charge.

1.5 moles of Ca will require 3F charge.

D. $FeO + H_2O \longrightarrow Fe_2O_3 + 2H^+ + 2e^-$

2 moles of FeO will require 2F charge.

1 mole of FeO will require 1 F charge.

67. *(a)* Electronegativity increases in period from left to right. The electronegativity values of given elements are,

Si = 1.8, C = 2.5

N = 3.0, O = 3.5

F = 4

Hence, the correct sequence is

Si < C < N < O < F.

68. *(c)* The correct IUPAC name and structure is,

$CH_3—\overset{\displaystyle 3^\circ}{\underset{\displaystyle CH_3}{\underset{|}{CH}}}—\overset{\displaystyle 3^\circ}{\underset{\displaystyle CH_3}{\underset{|}{CH}}}—CH_3$

2, 3-dimethylbutane

[2 tertiary (3°) carbon]

69. *(a)* Fehling reagent comprises of two solutions.

Fehling $A \rightarrow$ Aqueous solution of $CuSO_4$.

Fehling $B \rightarrow$ Alkaline sodium potassium tartrate (Rochelle's salt).

70. *(d)* Formula for calculating activation energy is given by

$$\log_{10}\left(\frac{k_2}{k_1}\right) = \frac{E_a}{2.303R}\left[\frac{1}{T_1} - \frac{1}{T_2}\right]$$

Here, k_1 is rate constant at T_1

k_2 is rate constant at T_2.

Hence, rate constant at two different temperatures should be known to calculate the activation energy of any chemical reaction.

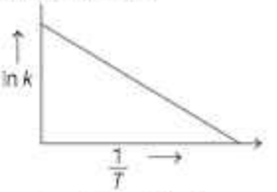
71. *(d)* According to Arrhenius equation,

$$k = A.e^{-E_a/RT} \Rightarrow \ln k = \ln A - \frac{E_a}{RT}$$

$$\ln k = \frac{-E_a}{R}\left(\frac{1}{T}\right) + \ln A$$

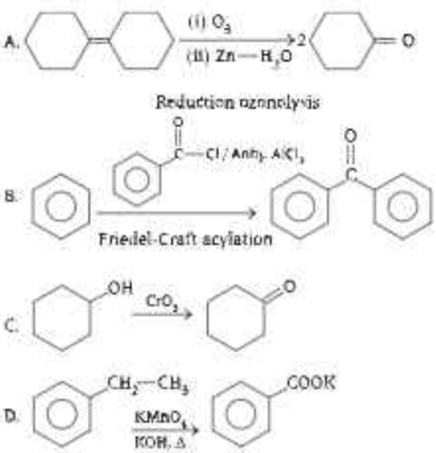
Compare it with $y = mx + c$

$\ln k \rightarrow y$ and $1/T \rightarrow x$

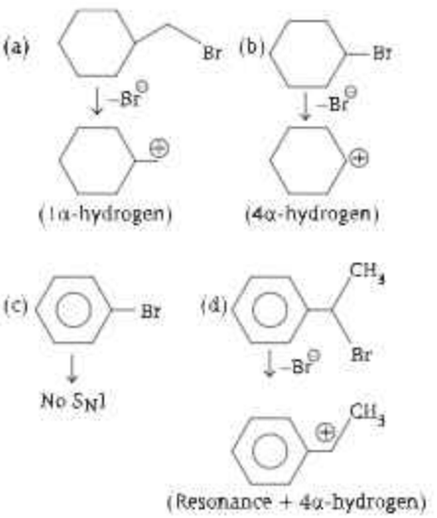


Hence, option (d) match the correct graph.

72. *(c)* The correct match is A-IV, B-I, C-II, D-III.



73. *(d)* Reactivity toward $S_N1 \propto$ stability of carbocation, and stability of carbocation depends on number of α-hydrogen.



Hence, (d) is correct option.

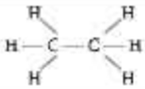
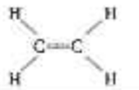
74. *(d)* $\underset{+2\ -1}{BaCl_2} + \underset{+1\ +6\ -2}{Na_2SO_4} \longrightarrow \underset{+2\ +6\ -2}{BaSO_4} + \underset{+1\ -1}{2NaCl}$

As there is no change in oxidation state of any element, hence it is not a redox reaction.

75. *(a)* Both statement I and statement II are true.

76. *(a)* Both statement I and statement II are true.

77. *(c)* The correct match is A-III, B-IV, C-II, D-I.

Molecule	Structure	(Type of bonds between two carbon atoms)
A. Ethane		1σ
B. Ethene		1 σ, 1 π
C. C_2	C≡C	2π (as per MOT)
D. Ethyne	H—C≡C—H	1 σ, 2π

78. *(b)* The relation between solubility and Henry's law constant is

$$\text{Solubility} \propto \frac{1}{K_H}$$

if K_H decreases then solubility increases.

K_H for gas *B* is lowest (2×10^{-5} kbar) and *A* has highest (145 k bar).

Hence, the order of solubility is given by

$$B > C > A.$$

79. *(a)* $E \propto \frac{Z^2}{n^2} \Rightarrow \frac{E_1}{E_2} = \frac{Z_1^2}{n_1^2} \times \frac{n_2^2}{Z_2^2}$

$He^+ \longrightarrow n_1 = 1$ and $Z_1 = 2$

$Be^{3+} \longrightarrow n_2 = 2$ and $Z_2 = 4$

$$\frac{E_1}{E_2} = \frac{2^2}{1^2} \times \frac{2^2}{4^2} = 1 \Rightarrow E_1 = E_2$$

Hence, it will be same as $-xJ$ for Be^{3+} ion.

80. *(c)* $E^\circ = +ve$ for Mn^{3+} / Mn^{2+}

$Mn = 25 \longrightarrow 3d^5 4s^2$

$Mn^{2+} \longrightarrow 3d^5$

$Mn^{3+} \longrightarrow 3d^4$

Mn^{2+} has half-filled ($3d^5$) stable configuration.

Thus, Mn^{3+} has very high tendency to gain an electron due to this it has high +ve value of electrode potential (E^c) while in other case, they does not gain electron easily.

Hence, the reason is change of d^4 to d^5-configuration.

81. *(c)* Glucose does not react with Schiff's reagent and $NaHSO_3$.

Glucose + Tollen's reagent ⟶ silver mirror

Glucose + HCN ⟶ cyanohydin formation

Glucose + NH_2OH ⟶ oxime formation

82. *(a)* The correct match is

A-II, B-III, C-IV, D-I.

83. *(b)* First ionisation enthalpy is the energy required to remove the first most loosely bound electron from an isolated atom in its ground state.

The electronic configuration is,

$Li \rightarrow 2s^1$, $Be \rightarrow 2s^2$, $B \rightarrow 2s^2, 2p^1$

$C \rightarrow 2s^2 2p^2$, $N \rightarrow 2s^2 2p^3$

The correct increasing order is,

$$Li < B < Be < C < N$$

84. *(b)* $NaOH + HCl \longrightarrow NaCl + H_2O$

Given, mass of NaOH = 1g

$$\text{Mole of NaOH} = \frac{1}{40}\text{ mol} = 0.025\text{ mol}$$

$$\text{Mole of HCl} = \frac{0.75 \times 25}{1000} = 0.01875\text{ mol}$$

Now finding the limiting reagent (LR)

$$\text{value} = \frac{\text{Given amount}}{\text{Stiochiometry coefficient}}$$

NaOH : HCl, $\frac{0.025}{1} > \frac{0.01875}{1}$

So, limiting reagent (LR) will be HCl.

Unreacted moles of NaOH

$= 0.025 - 0.01875 = 0.00625$ mol

Unreacted mass $= 0.00625 \times 40$ g

$= 0.25$ g

$= 250$ mg

85. *(c)* For the given reaction,

$2A \rightleftharpoons B + C$, $K_C = 4 \times 10^{-3}$

At a given time

$[A] = [B] = [C] = 2 \times 10^{-3}$ M

$$Q_C = \frac{[B][C]}{[A]^2} \Rightarrow Q_C = 1$$

As $Q_C > K_C$, reaction has tendency to go in backward direction.

86. *(a)*

Basic radical	Group number	
(B) Cu^{2+}	II	Group number increases
(A) Al^{3+}	III	
(D) Co^{2+}	IV	
(C) Ba^{2+}	V	
(E) Mg^{2+}	VI	

Hence, the correct order is

B, A, D, C, E.

87. *(d)* The reaction involves are,

$3R—OH + PCl_3 \longrightarrow 3RCl + H_3PO_3(A)$

$R—OH + PCl_5 \longrightarrow R—Cl + HCl + POCl_3(B)$

Hence, products *A* and *B* are

$A = H_3PO_3$; $B = POCl_3$

88. *(b)* Given, current (i) = 9.6487 A

1F = 96487 C

$t = 100$ s

$$w = \frac{E(it)}{96487} \Rightarrow w = \frac{\frac{63}{2} \times 9.6487 \times 100}{96487}$$

$w = 0.315$ g of Cu

89. *(a)* Osmotic pressure is given by,

$$\pi = CRT$$

Plot is of between osmotic pressure and concentration, i.e. π *vs* *C*.

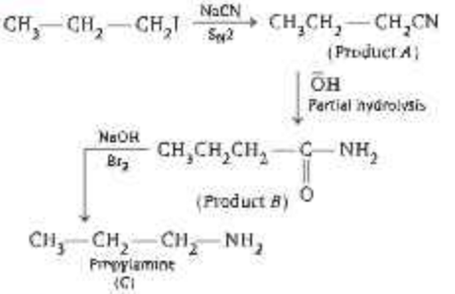
On comparing with $y = mx + C$

Slope $m = RT$

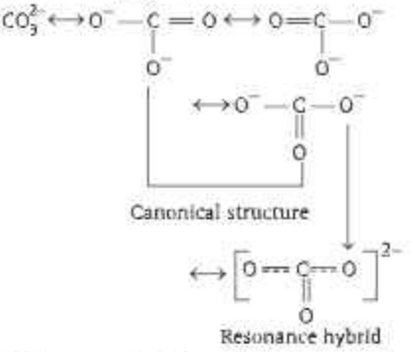
$25.73 = 0.083 \times T$

$T = 310\text{ K} = 37^\circ\text{C}$

90. *(a)* The complete reaction is as follow,



91. *(d)* Three canonical form can be drawn for CO_3^{2-} ion.



Other corrected forms are

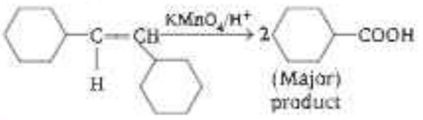
(a) Ozone has only two resonating structure.

(b) BF_3 has zero dipole moment.

(c) Dipole moment of NH_3 is greater than NF_3.

92. *(a)* Both statement I and statement II are true.

93. *(b)* The complete reaction is as follows,



94. *(a)* $Ce^{4+} = [Xe]\,4f^0\,5d^0$

$Yb^{2+} = [Xe]\,4f^{14}\,5d^0$

Both ions does not contain any unpaired electrons.

Hence, they are diamagnetic.

95. *(d)* $\underset{(2.8\times10^{-3}\,M)}{2NO(g)} \rightleftharpoons \underset{(3\times10^{-3}\,M)}{N_2(g)} + \underset{(4.2\times10^{-3}\,M)}{O_2(g)}$

$$K_C = \frac{[N_2][O_2]}{[NO]^2}$$

$$= \frac{3\times10^{-3}\times4.2\times10^{-3}}{(2.8\times10^{-3})^2} = 1.60 \quad \ldots (i)$$

$$\underset{t=t_{eq}\ (C-C\alpha)}{2NO(g)} \rightleftharpoons N_2(g) + O_2(g)$$

$$\frac{C\alpha}{2} \quad \frac{C\alpha}{2}$$

$$K_C = \frac{(C\alpha)^2}{4C^2(1-\alpha)^2}$$

$$\left(\frac{\alpha}{1-\alpha}\right)^2 = 4\times1.6 \text{ [from Eq. (i) } K_C = 1.60]$$

$$\frac{\alpha}{1-\alpha} = 2\times1.26 \Rightarrow \alpha = 0.717$$

96. *(b)*

Element	%	mole	Simple ratio
A	32	$\frac{32}{64} = \frac{1}{2}$	1
B	20	$\frac{20}{40} = \frac{1}{2}$	1
C	46	$\frac{48}{32} = \frac{3}{2}$	3

Empirical formula = ABC_3

97. *(b)* In reversible isothermal process, work done is,

$$W = -2.303nRT\log_{10}\left(\frac{p_1}{p_2}\right)$$

$$= -2.303\times1\times2\times298\log_{10}\left(\frac{20}{10}\right)$$

$$= -413.14 \text{ calories}$$

98. *(d)* Dilute H_2SO_4 do not act as oxidising agent which prevent oxidation of Fe^{2+} into Fe^{3+} while concentrated H_2SO_4 will oxidise Fe^{2+} into Fe^{3+}.

99. *(a)* Given, rate is quadruples, i.e. $r_2 = 4r_1$

We know, $\frac{r_2}{r_1} = 4 = \frac{k_2}{k_1}$

Now, $\log_{10}\left(\frac{k_2}{k_1}\right) = \frac{E_a}{2.303R}\left[\frac{1}{T_1} - \frac{1}{T_2}\right]$

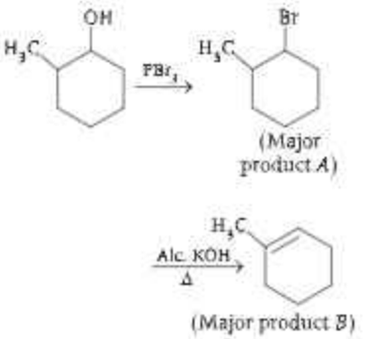
[where, $T_1 = 27°C = 300K$,

$T_2 = 57°C = 330K$]

$$\log_{10}(4) = \frac{E_a\times1000}{2303\times8.314}\left(\frac{1}{300} - \frac{1}{330}\right)$$

$E_a = 38.04$ kJ/mol

100. *(a)* The complete reaction is as follow,



Botany

101. *(b)* Lecithin is a type of phospholipid which is a class of lipids. These are the major components of cell membranes. Lecithin specifically contains a phosphate group, a glycerol backbone, two fatty acid chains and a choline molecule.

102. *(c)* The dark reaction of photosynthesis, also known as biosynthetic phase, occurs in the stroma of chloroplasts. It does not require direct light, instead it utilises the products of the light dependent reaction, i.e. ATP and NADPH, along with CO_2, to produce glucose.

These molecules provide the necessary energy and carbon source for the synthesis of glucose during the dark reaction of photosynthesis.

Therefore, the correct option is (c).

103. *(b)* During metaphase, the chromosomes are aligned at the metaphase plate, which is the equitorial plane of the cell. At this stage, spindle fibres, also known as microtubules, attach to the kinetochores, which are specialised protein structures located at the centromeres of the chromosomes.

These spindle fibres help to align the chromosomes properly along the metaphase plate. It also ensures that each chromatid will be pulled to opposite poles of the cell during anaphase.

104. *(a)* In grasses (monocots), certain adaxial epidermal cells along the veins modify themselves into large, empty, colourless cells. These are called bulliform cells. When the bulliform cells in the leaves have absorbed water and become turgid, the leaf surface is exposed. When they are flaccid due to water stress, they make the leaves curl inwards to minimise water loss.

105. *(a)* Component *c* represents guard cells.

Guard cells are kidney-shaped, specialised cells in plants that control the opening and closing of the stomata. The outer walls of guard cells are thin and inner walls are highly thickened.

Rest components can be identified as

A-**Epidermal cells** are parenchymatous with a small amount of cytoplasm lining the cell wall and a large vacuole.

B-**Subsidiary cells** are specialised epidermal cell, in the vicinity of the guard cells.

D-**Stomatal pores** are tiny openings on the surface of leaves that control gaseous exchange in plants.

106. *(c)* Statements B and C are correct regarding the piece of DNA carrying only gene of interest, transferred into an alien organism.

Rest statements are incorrect and these can be corrected as

- DNA typically requires host cellular machinery to replicate. It cannot multiply independently in progeny cells without the help of the host organism.
- When the DNA integrates into the host genome, it becomes an integral part of the chromosome.
- While the transferred DNA indeed have the ability to replicate if it contains appropriate regulatory sequences, replication alone does not guarantee its inheritance or integration into the host genome.

107. *(c)* Statement I is true, but statement II is false. The incorrect statement can be corrected as

The *Bt* toxin protein exists as inactive protoxins in *B. thuringiensis*, but once an insect ingest the inactive toxin, it is converted into an active form of toxin due to the alkaline pH of the gut.

108. *(d)* The International Union for Conservation of Nature and Natural Resources (IUCN) is a global organisation dedicated to assessing the conservation status of species and ecosystems. It releases lists of endangered species in Red Data Book. The IUCN Red List of threatened species is the most comprehensive source of information on the list of extinction for animal, plant and fungus species.
As of 2023, the IUCN Red List includes more than 157,100 species, with more than 44,000 threatened with extinction.

109. *(c)* Based on the given figure of a seed, the part destined to form the root when the seed germinates is the embryonic root or radicle *(C)*.
It is the first part of the embryo to develop into the plant's root system. The radicle grows downward into the soil, anchoring the seedling and allowing it to absorb water and nutrients.
Rest parts can be identified as
A-**Coleoptile** is a cylindrical, rigid, protective sheath that covers the first leaf and shoot apex of a grass seedling.
B-**Plumule** is the embryonic shoot of a plant.
D-**Coleorhiza** is a protective layer of tissue that surrounds the embryonic root or radicle of monocot seeds.

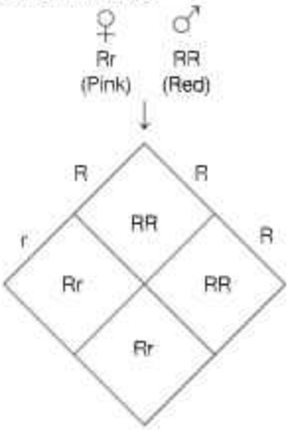
110. *(c)* A-III, B-I, C-IV, D-II

- ***Clostridium butylicum*** is a bacterium known for its ability to produce butyric acid through fermentation.
- *S. cerevisiae* is a yeast commonly used in fermentation process, such as brewing and baking, where it produces ethanol as a metabolic byproduct.
- ***Trichoderma polysporum*** is a fungus that produces cyclosporin-A, a potent immunosuppressive agent.
- ***Streptococcus* sp.** produces streptokinase, a thrombolytic agent.

111. *(d)* Both of the given figures *A* and *B* represent perigynous condition. In this condition gynoecium is situated in the centre and other parts of the flower are located on the rim of the thalamus almost at the same level.
The ovary in these flowers is said to be half inferior, e.g. plum, rose, peach, etc.

112. *(c)* Auxin is a plant hormone that regulates various aspects of plant development. They are commonly used by gardeners to prepare weed free lawns, it does not affect mature monocotyledonous plants (grass).
Auxin primarily promotes the abscission of mature leaves, particularly in broad leaf weeds, i.e. dictoyledonous weeds.
Rest other options are correct regarding auxin, but not relevant to the scenario described.

113. *(b)* The inheritance of flower colour in snapdragon shows incomplete dominance.
The cross between pink flower and red flower is as follows



Genotypic ratio—(Red) : Pink
1 1
Phenotypic ratio—RR : Rr
1 : 1
Thus, the expected progeny will be red as well as pink flowered plants.

114. *(b)* The morphology of the mycelium, mode of spore formation and fruiting bodies form the basis for the division of the kingdom into various classes. It does not depend on the mode of nutrition. Fungi are heterotrophs, they can not make their own food.

115. *(c)* Permease is a type of transporter protein that facilitates the movement of molecules such as lactose, across the cell membrane.
Other options can be explained as
In *E. coil*, it is coded by gene *Y*.
Inside the bacterial cell, the enzyme β-galactosidase catalyses the hydrolysis of lactose into its constituent monosaccharides, i.e., glucose and galactose.
This enzymatic reaction enables the bacterium to utilise lactose as a carbon source for energy and growth.

- Acetylase is involved in acetylation reactions. It speeds up the production of acetic acid esters.
- Polymerase is an enzyme involved in DNA replication and RNA transcription.

116. *(b)* To determine the genotype of the black seed plant, one needs to perform a test cross by crossing it with a plant that has the homozygous recessive genotype (bb).
It can be explained through the following crosses
Case I If parent is homozygous dominant.
Gametes from homozygous dominant parent (BB)

Gametes from recessive parent (bb)		B	B
	b	Bb	Bb
	b	Bb	Bb

All of the progeny will be of dominant phenotype.
Case II If parent is heterozygous dominant.
Gametes from heterozygous parent (Bb)

Gametes from recessive parent (bb)		B	b
	b	Bb	bb
	b	Bb	bb

It will results into 1 : 1 ratio of black and white seed.

117. *(d)* Statement I is false, but Statement II is true. Incorrect statement can be corrected as
Both parenchyma and collenchyma cells are living tissue. The collenchyma occurs in layers below the epidermis in most of the dicotyledonous plants. It consists of cells which are more thickened at the corners due to the deposition of cellulose, hemicellulose and pectin.

118. *(d)* Each turn of Calvin cycle requires a certain number of molecules of ATP and NADPH to provide energy and reducing power, respectively, for the various chemical reactions involved.
For every CO_2 molecule entering the Calvin cycle, 3 molecules of ATP and 2 molecules of NADPH are required.

119. *(d)* A transcription unit is a segment of DNA that takes part in transcription. It has three components.

(i) **Promoter** It is located towards 5' end of the structural gene and provides binding site for RNA polymerase.

(ii) **Structural gene** It codes for an RNA.

(iii) **Terminator** : It is located towards 3'-end of the coding strand and usually defines the end of the process of transcription.

120. *(a)* Statements A, C, D and E are the correct explanation for the species richness in tropical regions because

- tropical latitudes have remained relatively undisturbed for millions of years and thus had a long evolutionary time for species diversification.
- these environments are less seasonal, relatively more constant and predictable. Such constant environments promote niche specialisation. It leads to a greater species diversity.
- more solar energy is available in the tropics, which contributes to higher productivity. This, in turn might contribute indirectly to greater diversity.

121. *(c)* In the Verhulst-Pearl logistic growth equation, K represents the carrying capacity of the environment for a particular species. Carrying capacity is the maximum population size that the environment can sustainably support over the long term. It is determined by factors such as the availability of resources, space and other environmental conditions.

122. *(c)* Competitive inhibition occurs when a molecule, known as the inhibitor, competes with the substrate for binding to the active site of the enzyme. In the case of succinic dehydrogenase, malonate competes with the substrate for binding to the active site of the enzyme. Malonate structurally resembles succinate, the substrate for succinate dehydrogenase, and can therefore bind to the enzyme's active site.

Rest options can be explained as

- **Cofactor inhibition** involves the inhibition of an enzyme by binding to its cofactor, not the active site.
- **Feedback inhibition** occurs when the end product of a metabolic pathway inhibits an enzyme earlier in the pathway.
- **Enzyme activation** refers to processes that increase the activity of an enzyme.

123. (b) Statements A, C, D, and E can be explained on the basis of Mendel's law of dominance.

Statement B contradicts Mendel's law of dominance because, in the F_2-generation, the recessive alleles can still be expressed if present in the genotype, although they may be masked in the F_1-generation due to dominance.

124. *(a)* A-III, B-II, C-IV, D-I

- Nucleolus is the site for active ribosomal RNA synthesis.
- Centrioles are the two cylindrical structures in the centrosome. Each centriole appears like a cartwheel.
- Leucoplasts are colourless plastids of various shape and size that perform function for storing nutrients.

Golgi apparatus is the important site for the formation of glycoproteins and glycolipids in eukaryotic cells.

125. *(d)* Statements B, C, D and E are correct and statement A is not correct, it can be corrected as

Vallisneria flowers are not very colourful and do not produce nectar because they are water-pollinated flowers. These flowers do not need to attract insects for pollination.

126. *(a)* A-III, B-II, C-IV, D-I.

- *Rhizopus* is commonly known as bread mould. It belongs to Class-Phycomycetes of kingdom of Fungi.
- *Ustilago* is commonly known as smut. It belongs to Class-Basidiomycetes of kingdom of Fungi.
- *Puccinia* is commonly known as rust fungi.

It belongs to Class-Basidiomycetes of kingdom of Fungi.

- *Agaricus* is commonly called mushroom. It belongs to Class-Basidiomycetes of kingdom Fungi.

127. *(b)* *Hin*d II is a restriction endonuclease that cuts DNA molecules at a specific point by recognising a sequence of 6 base pairs. This specific base sequence is known as the recognition sequence. *Hin*d II specifically recognises 5'–AAGCTT - 3' -sequence.

128. *(a)* An actinomorphic flower is one that can be divided into symmetrical halves in any plane, typically through the center, resulting in multiple lines of symmetry. In the given example, *Datura* flower shows actinomorphic symmetry.

In *Cassia*, *Pisum* and *Sesbania*, zygomorphic flowers are seen.

129. *(b)* The type of conservation in which threatened animals and plants are taken out from their natural habitat and placed in special setting where they can be protected and given special care is called *ex situ* conservation. It is a type of biodiversity conservation.

The biodiversity conservation is a broad term that encompasses various conservation strategies, including both *in situ* and *ex situ* conservation.

130. *(a)* Both statement I and statement II are true.

131. *(c)* Dedifferentiation is a cellular process in plants where differentiated cells revert to a less differentiated or stem cell-like state. This process allows the cells to regain their ability to divide and re-enter the cell cycle.

In the case of the formation of interfascicular cambium from fully developed parenchyma cells, these parenchyma cells undergo dedifferentiation to become meristematic cells that are capable of division. These meristematic cells then differentiate into cambial cells, which give rise to secondary growth in plants.

Other options can be explained as

- Differentiation refers to the process by which undifferentiated cells become specialised into specific cell types.
- Redifferentiation is a process by which dedifferentiated cells regain their specialised characteristics.

132. *(a)* Totipotency refers to the capacity of a single plant cell to give rise to an entire new plant. Totipotent cells have the ability to regenerate into all types of cell and tissues. It plays crucial role in various aspects of plant biology, including regeneration, tissue culture and plant propagation.

Rest options can be explained as

- Micropropagation is a technique used to produce large numbers of genetically identical plants from a single parent plant using tissue culture methods.

- Somatic hybridisation involves the fusion of protoplasts from different plant species to produce hybrids with combined traits.

133. *(c)* A-III, B-IV, C-I, D-II.
- Two or more alternative forms of gene are called alleles.
- Cross of F_1-progeny with homozygous recessive parent is a test cross.
- Cross of F_1-progeny with any of the parents is a back cross.
- Number of chromosome sets in plant is called ploidy.

134. *(a)* Zinc is a cofactor for the proteolytic enzyme carboxypeptidase. A cofactor is a non-protein part of an enzyme that is made up of metal ions. In carboxypeptidase, the zinc cofactor is located in the active site of the enzyme and is essential for its catalytic ativity.
- Niacin is associated with coenzyme NAD and NADP.
- Haem is the prosthetic group in peroxidase and catalase.
- Flavin as a cofactor catalyses the oxidation of nucleophilic nitrogen, sulphur and phosphate.

135. *(d)* The accelerated rates of sepcies extinctions that the world is facing now are largely due to human activities.
These are four major causes ('The Evil Quartet' is the sobriquet used to describe them)
(i) Habitat loss and fragmentation
(ii) Overexploitation
(iii) Alien-species invasions
(iv) Co-extinctions
Hence, the correct option is (d).

136. *(a)* A-IV, B-II, C-I, D-III.
- In China rose monoadelphous androecium is present.
- Diadelophous androecium is found in pea plant.
- Polyadelphous androecium is found is citrus.
- Epiphyllous androecium is found in lily.

137. *(a)* A-IV, B-I, C-II, D-III.
- GLUT-4 is a glucose transporter protein found in the membrane of certain cells.
- Insulin is a hormone produced by the pancreas that regulates glucose metabolism.
- Trypsin is an enzyme involved in the digestion of proteins.
- Collagen is a structural protein found in the extracellular matrix of various tissues.

138. *(c)* In the tricarboxylic acid cycle, most of the steps involve the oxidation of substrates, leading to the generation of reducing equivalents, i.e., NADH and $FADH_2$ and ATP.
However, the conversion of succinyl Co-A to succinic acid by the enzyme succinyl Co-A synthetase, does not directly involve oxidation.
Rest other conversions involve oxidation.

139. *(b)* A-II, B-I, C-IV, D-III.
- The citric acid cycle occurs in the mitochondrial matrix.
- The glycolysis occurs in the cytoplasm of the cell.
- The electron transport system takes place in the inner mitochondrial membrane.
- The proton gradient is established across the intermembrane space of mitochondria.

140. *(b)* A-III, B-IV, C-I, D-II.
- Frederick Griffith's in a series of experiment with *Streptococcus pneumoniae* witness miraculous transformation in the bacteria.
- The elucidation of *lac* operon was a result of a close association between geneticist, Francois Jacob and a biochemist, Jacque Monod.
- Har Gobind Khorana developed chemical method to define combination of bases in genetic code.
- Meselson and Stahl provide proof of the semi-conservative mode of DNA replication.

141. *(c)* Statement I is true, but Statement II is false. It can be corrected as
Photorespiration does not occur in C_4 plants as they lack RuBisCO in mesophyll cells. Additionally, bundle sheath cells in C_4-plants do not show photorespiration due to the Kranz anatomy.

142. *(a)* The correct description about the given figure is given in option (a).
Wind pollinated flowers possess well exposed stamens. This is an adaptation to maximise pollen dispersal by wind. Having well exposed stamens ensures that the pollen is readily available for dispersal.

143. *(a)* A-II, B-IV, C-I, D-III.
- In rose, gynoecium is said to be perigynous.
- In pea, placenta forms a ridge along the ventral suture of the ovary and the ovules are borne on this ridge forming two rows. This type of placentation is known as marginal.
- In twisted aestivation, margin of the appendage overlaps that of the next one, e.g. cotton.
- In mango, fruit is drupe.

144. *(c)* Statements A, C, D and E are correct w.r.t the membets of Phaeophyceae. Statement B is not correct, it can be corrected as
In members of Phaeophyceae, sexual reproduction can be isogamous, anisogamous and oogamous. They produce motile gametes that are biflagellate, with two laterally attached flagella.

145. *(c)* The NPP of the first trophic level is the GPP of the second trophic level and the NPP of the second trophic level is GPP of the third trohpic level.
According to the 10% law, 90% is the respiratory loss at each trophic level.
$NPP = GPP - R$
At the first trophic level NPP is $100x$ which is GPP of second trophic level.
At second trophic level NPP is 10% of GPP $= 10x$. Since, the GPP of the 3rd trophic level is the same as the NPP of the second trophic level, it is equal to $10x$.

146. *(d)* Statement in option (d) is correct regarding the process of replication in *E. coli*. It can be explained as
In living cells, such as E. coli, the process of replication requires a set of catalysts .or enzymes. The main enzyme is referred to as DNA-dependent DNA polymerase. These enzymes use a DNA template to catalyse the polymerisation of deoxynucleotides. The enzyme DNA-dependent DNA polymerase catalyses polymerisation only in one direction, i.e. $5' \rightarrow 3'$.

147. *(c)* In somatic hybridisation, protoplasts of two varieties of plants are fused. During this process, protoplasts are isolated and brought together and fused using techniques such as chemical treatment or electric fields. This fusion combines the genetic material from the two parent plants, leading to the formation of somatic hybrids.

148. *(b)* Spraying sugarcane crop with gibberellin increases the length of the stem, thus increasing the yield by as much as 20 tonnes per acre.

Gibberellins are plant hormones that promote stem elongation, cell division and cell expansion. Applying gibberellin to sugarcane crops, stimulate the elongation of the internodes. This increased stem length allows for more space for sugar accumulation and ultimately leads to higher yields.

Other options can be corrected as

- Auxin primarily regulates plant growth by promoting cell elongation and is not responsible for stem elongation in sugarcane.
- Cytokinin is involved in cell division and lateral bud development, whereas abscisic acid acts as stress hormone.

149. *(b)* A-III, B-I, C-IV, D-II.

- Robert May places the global species diversity at about 7 million.
- Alexander von Humboldt provided species-area relationship.
- Paul Ehrlich used an analogy "rivet popper hypothesis" to explain the role of species in the ecosystem.
- David Tilman performed long term ecosystem experiments using outdoor plots.

150. *(b)* Chloroplasts are organelles found in plant cells, responsible for photosynthesis. They contain their own genome, which is typically a circular molecule of DNA. This DNA is double stranded.

The circular nature of chloroplast DNA is a characteristic feature shared with bacterial DNA, reflecting the endosymbiotic theory.

Zoology

151. *(c)* A-III, B-I, C-II, D-IV.

- Common cold is caused by rhinoviruses.
- Haemozoin is a chemical released due to the rupturing of RBC by *Plasmodium*.
- Widal test is used to diagnose typhoid.
- Allergy is caused by allergens like dust, mites.

152. *(d)* A-III, B-IV, C-I, D-II.

- Cocaine is derived from coca plant, i.e. *Erythroxylum*.
- Heroin is an opioid obtained from latex of poppy plant, *Papaver somniferum*.
- Morphine is an effective sedative used for treating patients who have undergone surgery.
- Marijuana is a cannabinoid obtained from the plant *Cannabis sativa*.

153. *(d)* A-III, B-I, C-IV, D-II.

- Fibrous joints are present in skull. They do not allow any movement.
- Cartilaginous joints are present in adjacent vertebrae. They provide limited movement.
- Hinge joint is present in knee and helps in locomotion.
- Ball and socket joints are present between humerus and pectoral girdle and provide rotational movement.

154. *(b)* Among the given options myasthenia gravis, rheumatoid arthritis and Systemic Lupus Erythematosus (SLE) are autoimmune disorders. The autoimmune disease is a type of disease in which the body's immune system attacks healthy tissues of the body. Other options can be explained as

Gout is an inflammatory disorder and muscular dystrophy is a progressive degeneration of skeletal muscles in which mutated genes interfere with healthy muscle function.

155. *(a)* Uterine fundus is not a component of Fallopian tube. It is the part of uterus that is farthest away from its opening, the cervix. On the other hand isthmus, infundibulum and ampulla are the parts of Fallopian tube.

156. *(c)* The flippers of the penguins and dolphins are the examples of convergent evolution. The convergent evolution can be defined as a process in which unrelated organisms evolve similar body forms, colourations and adaptations. However, they do not belong to a common ancestor.

157. *(c)* A-III, B-IV, C-I, D-II.

- α-1 antitrypsin is used for treatment of emphysema.
- *cry* I Ab gene controls corn borer.
- *cry* I Ac controls cotton bollworm.
- Enzyme replacement therapy can be used as treatment in ADA deficiency.

158. *(b)* In the given figure of pBR322, *X* represents *'ori'* region, while *Y* represent *'rop'* region. The *'ori'* site is the origin of replication that controls the copy number of the linked DNA. Whereas *rop* codes for proteins which are involved in the process of replication of the plasmid.

159. *(a)* Both A and R are correct and R is the correct explanation of A.

Doctors recommend breast-feeding during initial period of infant growth because it contains IgA antibodies that provide passive immunity to the new born baby.

160. *(c)* The 'Ti-plasmid' of *Agrobacterium tumefaciens* stands for tumour inducing plasmid. It induces crown gall disease in plants by transferring crucial regions from the plasmid.

161. (b) A-II, B-I, C-IV, D-III.

- *Pleurobrachia* is a member of phylum-Ctenophora.
- Radula is a rasping or feeding organ present in phylum-Mollusca.
- Stomochord is rudimentary structure similar to notochord found in collar region of members of phylum-Hemichordata.
- Air bladder is found in Osteichthyes, which regulates buoyancy.

162. (d) The correct sequence of human evolution from past to recent is as follows

Homo habilis → *Homo erectus* → *Homo neanderthalensis* → *Homo sapiens*

Homo habilis appeared around 3.2 mya. They were capable of making sharpened tools and their cranial capacity was 650-800 cc.

Homo erectus appeared around 1.5 mya. They had erect body posture and cranial capacity was 900 cc.

Homo neanderthalensis appeared around 40,000 to 1,00,000 years ago. They had cranial capacity of 1400 cc.

Homo sapiens is the modern man evolved somewhere between 75,000 to 10,000 years ago. Their cranial capacity is 1450 cc.

163. *(d)* Glucagon is a peptide hormone.

It is produced by α-cells of the pancreas. It raises the concentration of glucose and fatty acids in the blood stream. On the other hand, cortisol,

testosterone and progesterone are the examples of steroid hormones.

164. *(b)* Anal cerci, a pair of jointed filamentous structures are present on 10th segment in both sexes of cockroach. These are sense organs that are sensitive to wind and also detect vibration.

165. *(d)* Among the given options, a constant gene pool will not affect the Hardy-Weinberg equilibrium. It was proposed by GH Hardy and W Weinberg independently in 1908. According to this, a stable population has constant allelic frequencies in sexually reproducing organisms from generations to generations. The factors which cause change in allelic frequencies in a population are recombination, mutation, gene migration, genetic drift and natural selection.

166. *(b)* A-III, B-IV, C-II, D-I.
- Pons connects different regions of the brain.
- Hypothalamus also have neuro secretory cells which secrete hormone.
- Medulla controls respiration and gastric secretion.
- Cerebellum provides additional space for neurons, regulates posture and balance.

167. *(c)* A-III, B-IV, C-I, D-II.
- Down's syndrome is caused due to trisomy of 21st chromosome.
- α-thalassemia is controlled by two closely linked genes HBA1 and HBA2 on chromosome 16 of each parent and occurs due to mutation or deletion of one or more of the four genes.
- β-thalassemia is controlled by a single gene HBB on chromosome 11 of each parent and occurs due to mutation of one or both genes.
- Klinefelter's syndrome is caused due to the presence of an additional copy of X chromosome resulting into a karyotype of 47, XXY.

168. *(a)* The strand of DNA which is copied into *m*RNA and subsequently translated, possesses complementary sequence to the *m*RNA. It is called as template strand. During transcription, RNA polymerase binds to the template strand and synthesis RNA. However, due to the complementarity rule, in RNA uracil is present in place of thyamine. Therefore, the correct product will be determined as follows :

3' TACATGGCAAATATCCATTCA5'

5' AUGUACCGUUUAUAGGUAAGU3'

169. *(d)* A is false, but R is true. A can be corrected as

LH and FSH both are secreted by anterior pituitary. In females, FSH acts upon ovarian follicles; while LH stimulates Leydig's cells of testis to secrete androgen.

170. *(d)* Among the given options, vaults are not the natural contraceptive methods instead, they are barrier methods. The vaults are disposable cup like structures made of rubber that are inserted into the vaginal tract to cover the cervix before coitus. They prevent the entry of sperms through cervix to prevent fertilisation.

171. *(d)* A-III, B-I, C-IV, D-II.
- Example of non-medicated IUD is lippes loop.
- Example of copper releasing IUD is Multiload 375.
- Example of hormone releasing IUD is LNG-20.
- Example of implants is progestogens.

172. *(b)* Among the given statements, only statement A is correct. Rest of the statements are incorrect and can be corrected as follows
- Poriferans and Platyhelminthes are acoelomate as they lack coelom.
- Aschelminthes are pseudocoelomates.

173. *(b)* (i) represents skeletal muscles which is present in triceps. These are voluntary muscles and show striations. (ii) represents smooth muscles which are involuntary in action and are present in stomach, whereas (iii) represents cardiac muscles which are exclusively found in heart.

174. *(a)* The correct sequence of pathway for conduction of an action potential through the heart is as follows

SA node → AV node → AV bundle → Bundle branches → Purkinje fibres

175. *(c)* A-II, B-IV, C-I, D-III.
- Lipase helps in digestion of ester bond.
- Nuclease helps in digestion of phosphodiester bond.
- Protease helps in digestion of peptide bond.
- Amylase helps in digestion of glycosidic bond.

176. (d) A-II, B-I, C-IV, D-III.
- Axoneme is seen in cilia and flagella.
- Centriole shows cartwheel appearance.
- Crista is found in mitochondria.
- Few chromosomes have secondary constrictions at a constant location. This gives the appearance of a small fragment called satellite.

177. *(c)* A-II, B-IV, C-I, D-III.
- In diakinesis, completion of terminalisation of chiasmata takes place.
- In pachytene, appearance of recombination nodule takes place.
- In zygotene, synaptonemal complex formation takes place.
- In leptotene, chromosomes look like thin threads.

178. *(b)* Among the given factors, high pO_2 and lesser H^+ concentration are favourable for the formation of oxyhaemoglobin in alveoli. The higher the partial pressure of oxygen. The more efficiently haemoglobin binds to O_2.

179. *(b)* A-III, B-I, C-II, D-IV.
- *Pterophyllum* is the scientific name for angel fish.
- *Myxine* is the scientific name for hag fish.
- *Pristis* is the scientific name for saw fish.
- *Exocoetus* is the scientific name for flying fish.

180. *(b)* A-IV, B-III, C-I, D-II.
- Typhoid is caused by *Salmonella typhimurium* (Bacteria).
- Leishmaniasis is caused by protozoan i.e. *Leishmania donovani*.
- Ringworm is caused by fungi belonging to genera-*Microsporum*.
- Filariasis is caused by *Wuchereria bancrofti* (Nematode).

181. *(c)* All the statements are correct except (c), which can be corrected as. Bio-reactors are used to produce large scale bacterial cultures.

182. *(b)* Both statement I and II are incorrect. These can be corrected as
- In the nephron, the decending limb of loop of Henle is permeable to water, but almost impermeable to electrolytes.

- The PCT is lined by simple cuboidal brush border epithelium which increases the surface area for reabsorption.

183. *(c)* Statement I is true, but statement II is false. Statement II can be corrected as
The hymen is often torn during first coitus. However, it may get torn due to a sudden fall or a jolt during horse-riding, cycling, gymnastics and dance movements, etc.

184. *(a)* A-II, B-IV, C-I, D-III.

• Expiratory capacity	Total volume + Expiratory reserve volume.
• Functional residual capacity	Expiratory reserve volume + Residual volume
• Vital capacity	Expiratory reserve volume + Tidal volume + Inspiratory reserve volume
• Inspiratory capacity	Tidal volume + Inspiratory reserve volume

185. *(d)* The correct sequence of different stages of cell division is as follows
Gap 1 phase → Synthesis phase → Gap 2 phase → Karyokinesis → Cytokinesis
In Gap 1 and Gap 2 phases, cell prepares itself for cell division; while actual division occurs in synthesis phase. Karyokinesis is division of nucleus that is followed by cytokinesis.

186. *(c)* Statement I is true, but Statement II is false. The incorrect statement can be corrected as
The inner membrane of chloroplast is relatively less permeable; as compared to mitochondria.

187. *(d)* A-III, B-I, C-IV, D-II.
- Birds and reptiles evolved in mesozoic era.
- Lower invertebrates evolved in Proterozoic era.
- Mammals evolved in Cenozoic era.
- Fish and amphibians evolved in Palozoic era.

188. *(d)* Statement I is false, but statement II is true. Statement I can be corrected as.
Gause's competitive exclusion theory states that two closely related species competing for the same resources cannot co-exist indefinitely.

189. *(c)* A-III, B-IV, C-I, D-II.
- Unicellular glandular epithelium is present in goblet cells of alimentary canal.
- Compound epithelium is present in moist surface of buccal cavity.
- Multicellular glandular epithelium is present in salivary glands.
- Endocrine glandular epithelium is present in pancreas.

190. *(a)* A-IV, B-II, C-III, D-I.
- The structures used for storing of food are crop.
- Ring of 6-8 blind tubules at junction of foregut and midgut is gastric caeca.
- Ring of 100-150 yellow coloured thin filaments at the junction of midgut and hindgut are called Malpighian tubules.
- The structures used for grinding the food are gizzard.

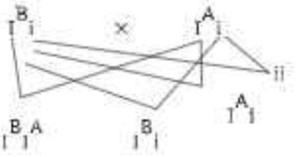
191. *(c)* Statement (c) is correct, while rest of the statements are incorrect. These can be corrected as follows.
- Juxta medullary nephrons are not present in columns of Bertini.
- Renal corpuscle of juxta medullary nephron lies in the inner cortical region.
- Juxta medullary nephrons are lesser in number than cortical nephrons.

192. *(d)* A-IV, B-III, C-I, D-II.
- In eukaryotes, RNA polymerase-III is responsible for transcription of *sn*RNAs, *t*RNA and 5*sr*RNA
- rho factor is responsible for termination of transcription.
- Splicing of exons is performed by *sn*RNPs.
- TATA box is present in promoter region of transcription unit.

193. *(c)* Statement I is true, but statement II is false. Statement II can be corrected as
The brain stem consist of the medulla oblongata, pons and midbrain.
Hence, option (c) is the correct answer.

194. *(a)* In the given question A, B, C and D are respectively FSH, leydig cells, Sertoli cells and Spermiogenesis. LH and FSH both are secreted by anterior pituitary. LH stimulates Leydig's cells to secrete androgens. FSH stimulates Sertoli cells that release some factors which help in spermiogenesis.

195. *(a)* If the blood group of father is B^+, mother is A^+ and child is O^+ their respective genotypes will be $I^Bi / I^Ai /$ ii



196. *(a)* Both Statement I and Statement II are true.

197. *(a)* The catalytic cycle of an enzyme action occurs in the following steps
Substrate binding to active site (E)
↓
Substrate enzyme complex formation (A)
↓
Chemical bonds of the substrate broken (D)
↓
Release of products (C)
↓
Free enzyme ready to bind with another substrate (B)
Thus, the correct answer is option (a).

198. *(b)* A-III, B-II, C-IV, D-I.
- P-wave–Depolarisation of atria.
- QRS complex–Depolarisation of ventricles.
- T-wave–Repolarisation of ventricles.
- T-P gap–Heart muscles are electrically silent.

199. *(d)* A-III, B-IV, C-I, D-II.
- Exophthalmicgoitre – Hypersecretion of thyroid hormone and protruding eye balls.
- Acromegaly – Excessive secretion of growth hormone.
- Cushing's syndrome – Excess secretion of cortisol, moon face and hyperglycemia.
- Cretinism – Hyposecretion of thyroid hormone and stunted growth.

200. *(c)* Statements B, D and E are true for non-chordates, while statements A and C are incorrect. These can be corrected as
- Pharynx is not perforated by gill slits in non-chordates.
- In non-chordates, the central nervous system is ventral and solid.